MATH 101 **86031**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Which pair satisfies h(x) = f(g(x))?

$h(x) = (3x + 3)^2 + 4$

$A.\quad f(x) = x^2 + 4,\ g(x) = 3x + 3$

$B.\quad f(x) = x^2 + 3,\ g(x) = 3x + 4$

$C.\quad f(x) = 3x^2 + 4,\ g(x) = x + 3$

$D.\quad f(x) = 3x + 3,\ g(x) = x^2 + 4$

1.

10 pts

2. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end.

$f(x) = \sqrt{x + 4} - 2$

2. (a)

2 pts

(b)

2 pts

(c)

1 pts

(d)

1 pts

(e)

3 pts

(f)

1 pts

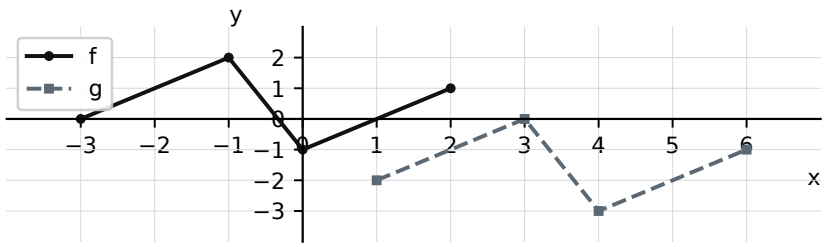
3. The solid graph is f and the dashed graph is g. Select the expression for g(x).

$A.\quad g(x) = f(x + 4) + 2$

$B.\quad g(x) = f(x - 4) - 2$

$C.\quad g(x) = f(x - 4) + 2$

$D.\quad g(x) = f(x + 4) - 2$



3.

10 pts

Test continues on the other side of the sheet.

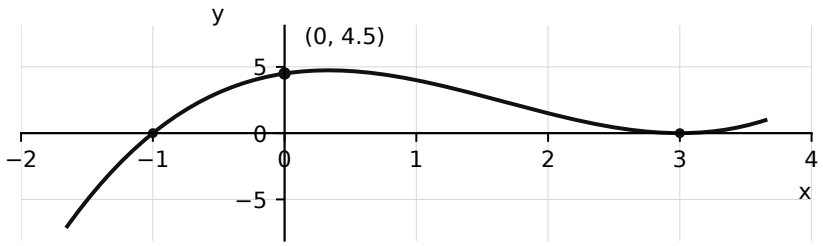
MATH 101 **86031**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form. 	4. 10 pts
5. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_2(x-2) + 1$	5. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
6. Find all zeros. $p(x) = x^3 - 2x^2 - 4x + 3$	6. 10 pts

Test continues on the next sheet.

MATH 101 **86031**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

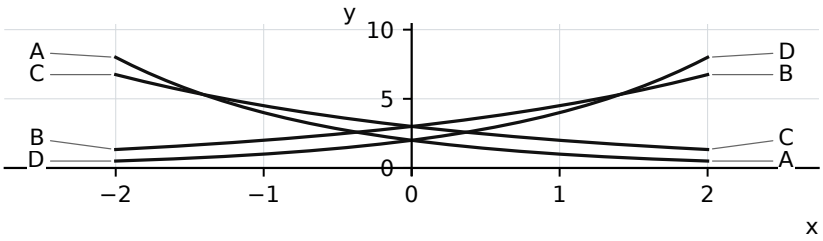
7. Match each function to graph A, B, C, or D. Each letter is used once.

(a) $3(\frac{2}{3})^x$

(b) $2(\frac{1}{2})^x$

(c) $2(2)^x$

(d) $3(\frac{3}{2})^x$



7. (a) 3 pts

(b) 2 pts

(c) 3 pts

(d) 2 pts

8. Solve exactly.

$\sqrt{x+3} = x-3$

8. 10 pts

9. Evaluate exactly. Here log means base 10.

(a) $\log(0.25) + \log(400)$

(b) $\frac{\log_6(27)}{\log_6(9)}$

9. (a) 5 pts

(b) 5 pts

10. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept.

$f(x) = \frac{(x+2)(x+3)}{(x-5)(x+2)}$

10. (a) 2 pts

(b) 2 pts

(c) 1 pts

(d) 1 pts

(e) 2 pts

(f) 2 pts

86031

TEST PAGE 1

1.

A

10 pts

2. (a)

$[-4, \infty)$ 2 pts

(b)

$[-2, \infty)$ 2 pts

(c)

$(0, 0)$ 1 pts

(d)

$(0, 0)$ 1 pts

(e)

$x \to \infty : f(x) \to \infty$ 3 pts

(f)

$x \to -4^+ : f(x) \to -2$ 1 pts

3.

B

10 pts

TEST PAGE 2

4.

$\frac{1}{2}(x+1)(x-3)^2$

10 pts

5. (a)

$(2, \infty)$ 2 pts

(b)

$(-\infty, \infty)$ 2 pts

(c)

$x = 2$ 1 pts

(d)

$(\frac{5}{2}, 0)$ 1 pts

(e)

$x \to 2^+ : f(x) \to -\infty$ 2 pts

(f)

$x \to \infty : f(x) \to \infty$ 2 pts

6.

$x = 3,\ -\frac{1}{2} + \frac{\sqrt{5}}{2},\ -\frac{\sqrt{5}}{2} - \frac{1}{2}$ 10 pts

TEST PAGE 3

7. (a)

C

3 pts

(b)

A

2 pts

(c)

D

3 pts

(d)

B

2 pts

8.

$x = 6$

10 pts

9. (a)

2 5 pts

(b)

$\frac{3}{2}$ 5 pts

10. (a)

$\mathbb{R} \setminus \{-2, 5\}$ 2 pts

(b)

$(-2,\ -\frac{1}{7})$ 2 pts

(c)

$x = 5$ 1 pts

(d)

$y = 1$ 1 pts

(e)

$(-3, 0)$ 2 pts

(f)

$(0,\ -\frac{3}{5})$ 2 pts

MATH 101 **13685**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Evaluate exactly. Here log means base 10.

(*a*) $\log(0.8) + \log(1250)$

(*b*) $\frac{\log_6(27)}{\log_6(9)}$

1. (a) 5 pts

(b) 5 pts

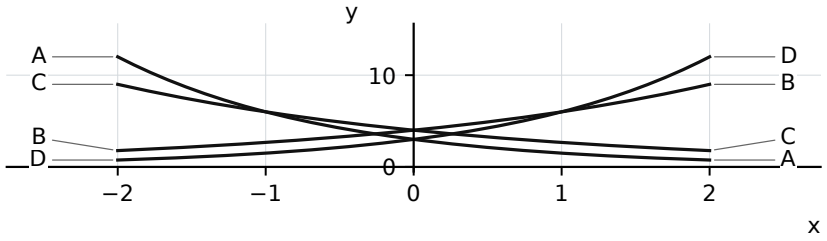
2. Match each function to graph A, B, C, or D. Each letter is used once.

(*a*) $4(\frac{2}{3})^x$

(*b*) $3(\frac{1}{2})^x$

(*c*) $3(2)^x$

(*d*) $4(\frac{3}{2})^x$



2. (a) 3 pts

(b) 2 pts

(c) 3 pts

(d) 2 pts

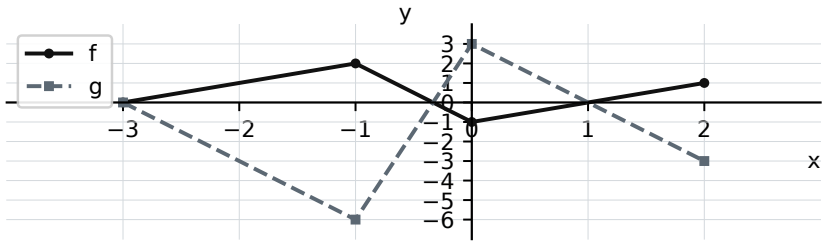
3. The solid graph is f and the dashed graph is g. Select the expression for g(x).

A. $g(x) = f(x) - 3$

B. $g(x) = -3f(x)$

C. $g(x) = 3f(x)$

D. $g(x) = f(-3x)$



3. 10 pts

Test continues on the other side of the sheet.

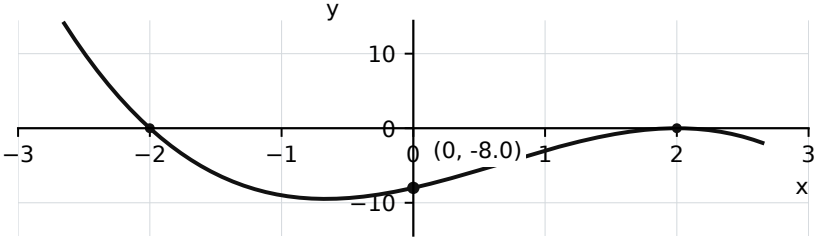
MATH 101 **13685**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_3(x-2) + 1$	4. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
5. Which pair satisfies h(x) = f(g(x))? $h(x) = (3x+3)^2 + 4$ $A.\quad f(x) = x^2 + 4,\ g(x) = 3x + 3$ $B.\quad f(x) = x^2 + 3,\ g(x) = 3x + 4$ $C.\quad f(x) = 3x^2 + 4,\ g(x) = x + 3$ $D.\quad f(x) = 3x + 3,\ g(x) = x^2 + 4$	5. 10 pts
6. Find all zeros. $p(x) = x^3 + 5x^2 + 7x + 2$	6. 10 pts
7. Solve exactly. $8^{x-2} = 4^{x+5}$	7. 10 pts
8. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form. 	8. 10 pts

Test continues on the next sheet.

MATH 101 **13685**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

9. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = \sqrt{x+9} - 2$	9. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts
10. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x+2)(x+3)}{(x-5)(x+2)}$	10. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts

13685

TEST PAGE 1

1. (a)

3 — 5 pts

(b)

$\frac{3}{2}$ — 5 pts

2. (a)

C — 3 pts

(b)

A — 2 pts

(c)

D — 3 pts

(d)

B — 2 pts

3.

B — 10 pts

TEST PAGE 2

4. (a)

$(2, \infty)$ — 2 pts

(b)

$(-\infty, \infty)$ — 2 pts

(c)

$x = 2$ — 1 pts

(d)

$(\frac{7}{3}, 0)$ — 1 pts

(e)

$x \to 2^+ : f(x) \to -\infty$ — 2 pts

(f)

$x \to \infty : f(x) \to \infty$ — 2 pts

5.

A — 10 pts

6.

$x = -2, -\frac{3}{2} - \frac{\sqrt{5}}{2}, -\frac{3}{2} + \frac{\sqrt{5}}{2}$ — 10 pts

7.

$x = 16$ — 10 pts

8.

$-1(x+2)(x-2)^2$ — 10 pts

TEST PAGE 3

9. (a)

$[-9, \infty)$ — 2 pts

(b)

$[-2, \infty)$ — 2 pts

(c)

$(-5, 0)$ — 1 pts

(d)

$(0, 1)$ — 1 pts

(e)

$x \to \infty : f(x) \to \infty$ — 3 pts

(f)

$x \to -9^+ : f(x) \to -2$ — 1 pts

10. (a)

$\mathbb{R} \setminus \{-2, 5\}$ — 2 pts

(b)

$(-2, -\frac{1}{7})$ — 2 pts

(c)

$x = 5$ — 1 pts

(d)

$y = 1$ — 1 pts

(e)

$(-3, 0)$ — 2 pts

(f)

$(0, -\frac{3}{5})$ — 2 pts

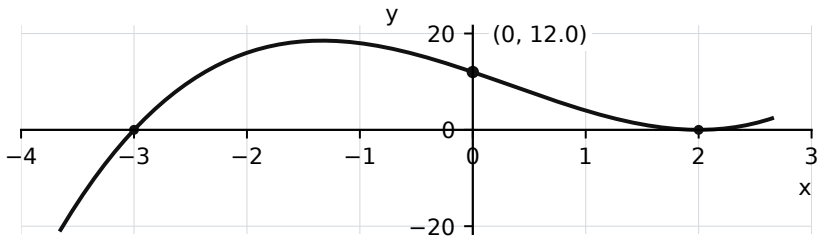
MATH 101 **87553**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form.  	1. 10 pts
2. Evaluate exactly. Here log means base 10. (a) $\log(0.05) + \log(200000)$ (b) $\frac{\log_6(27)}{\log_6(9)}$	2. (a) 5 pts (b) 5 pts
3. Solve exactly. $\log_3(x-3) + \log_3(2) = 2$	3. 10 pts
4. Find all zeros. $p(x) = x^3 + 3x^2 - 2$	4. 10 pts
5. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_4(x-2) + 1$	5. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts

Test continues on the other side of the sheet.

MATH 101 **87553**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

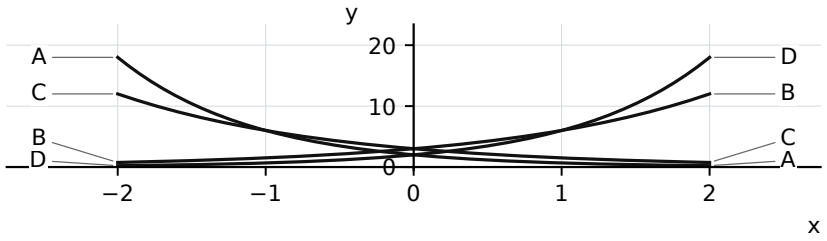
6. Match each function to graph A, B, C, or D. Each letter is used once.

$(a)\quad 3(\frac{1}{2})^x$

$(b)\quad 2(\frac{1}{3})^x$

$(c)\quad 2(3)^x$

$(d)\quad 3(2)^x$



6. (a)

3 pts

(b)

2 pts

(c)

3 pts

(d)

2 pts

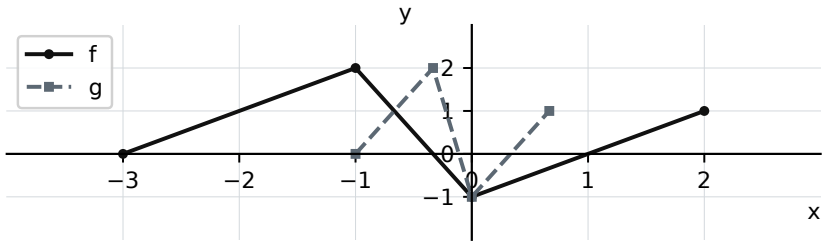
7. The solid graph is f and the dashed graph is g. Select the expression for g(x).

A. $g(x) = f(x-3)$

B. $g(x) = f(3x)$

C. $g(x) = f(x/3)$

D. $g(x) = 3f(x)$



7.

10 pts

8. Which pair satisfies h(x) = f(g(x))?

$h(x) = (3x+3)^2 + 4$

A. $f(x) = x^2 + 4,\ g(x) = 3x + 3$

B. $f(x) = x^2 + 3,\ g(x) = 3x + 4$

C. $f(x) = 3x^2 + 4,\ g(x) = x + 3$

D. $f(x) = 3x + 3,\ g(x) = x^2 + 4$

8.

10 pts

Test continues on the next sheet.

MATH 101 **87553**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

9. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x+2)(x+3)}{(x-5)(x+2)}$	9. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
10. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = \sqrt{x+16} - 2$	10. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts

87553

TEST PAGE 1

1.

$1(x+3)(x-2)^2$

10 pts

2. (a)

4 5 pts

(b)

$\frac{3}{2}$ 5 pts

3.

$x=\frac{15}{2}$ 10 pts

4.

$x=-1,\ -1+\sqrt{3},\ -\sqrt{3}-1$ 10 pts

5. (a)

$(2,\infty)$ 2 pts

(b)

$(-\infty,\infty)$ 2 pts

(c)

$x=2$ 1 pts

(d)

$(\frac{9}{4},0)$ 1 pts

(e)

$x\to 2^+ : f(x)\to -\infty$ 2 pts

(f)

$x\to\infty : f(x)\to\infty$ 2 pts

TEST PAGE 2

6. (a)

C

3 pts

(b)

A

2 pts

(c)

D

3 pts

(d)

B

2 pts

7.

B

10 pts

8.

A

10 pts

TEST PAGE 3

9. (a)

$\mathbb{R}\setminus\{-2,5\}$ 2 pts

(b)

$(-2,\ -\frac{1}{7})$ 2 pts

(c)

$x=5$ 1 pts

(d)

$y=1$ 1 pts

(e)

$(-3,0)$ 2 pts

(f)

$(0,\ -\frac{3}{5})$ 2 pts

10. (a)

$[-16,\infty)$ 2 pts

(b)

$[-2,\infty)$ 2 pts

(c)

$(-12,0)$ 1 pts

(d)

$(0,2)$ 1 pts

(e)

$x\to\infty : f(x)\to\infty$ 3 pts

(f)

$x\to -16^+ : f(x)\to -2$ 1 pts

MATH 101 **74618**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end.

$f(x) = 2\sqrt{x+4} - 6$

1. (a) 2 pts

(b) 2 pts

(c) 1 pts

(d) 1 pts

(e) 3 pts

(f) 1 pts

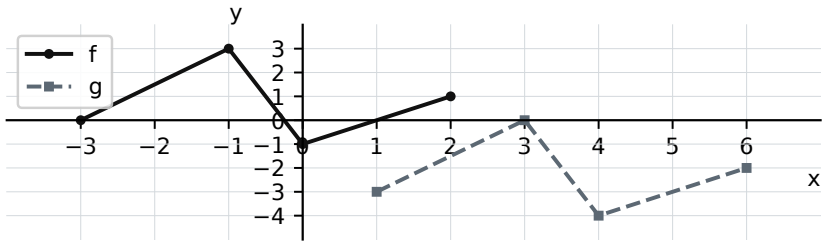
2. The solid graph is f and the dashed graph is g. Select the expression for g(x).

$A.\quad g(x) = f(x-4) + 3$

$B.\quad g(x) = f(x+4) - 3$

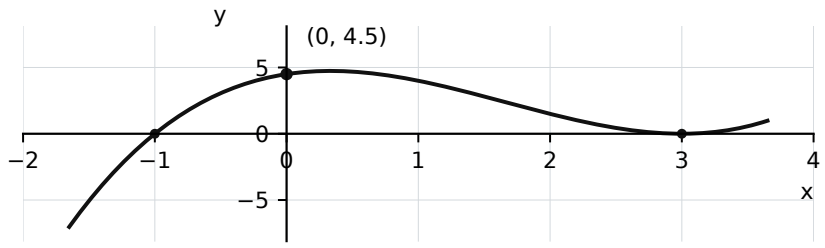
$C.\quad g(x) = f(x-4) - 3$

$D.\quad g(x) = f(x+4) + 3$



2. 10 pts

3. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form.



3. 10 pts

Test continues on the other side of the sheet.

MATH 101 **74618**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept.

$f(x) = \frac{(x+3)(x+4)}{(x-6)(x+3)}$

4. (a) 2 pts

(b) 2 pts

(c) 1 pts

(d) 1 pts

(e) 2 pts

(f) 2 pts

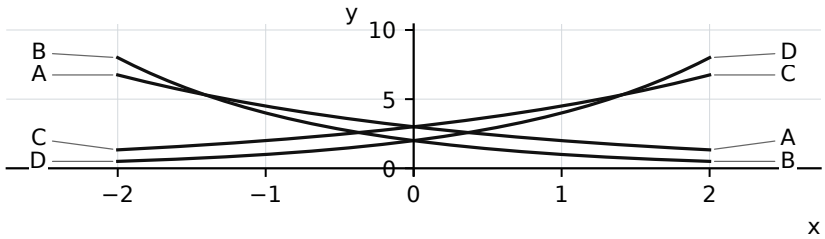
5. Match each function to graph A, B, C, or D. Each letter is used once.

$(a)\quad 3(\frac{3}{2})^x$

$(b)\quad 3(\frac{2}{3})^x$

$(c)\quad 2(2)^x$

$(d)\quad 2(\frac{1}{2})^x$



5. (a) 3 pts

(b) 2 pts

(c) 3 pts

(d) 2 pts

6. Which pair satisfies h(x) = f(g(x))?

$h(x) = (4x+4)^2 + 5$

$A.\quad f(x) = x^2 + 5,\ g(x) = 4x + 4$

$B.\quad f(x) = 4x + 4,\ g(x) = x^2 + 5$

$C.\quad f(x) = x^2 + 4,\ g(x) = 4x + 5$

$D.\quad f(x) = 4x^2 + 5,\ g(x) = x + 4$

6. 10 pts

Test continues on the next sheet.

MATH 101 **74618**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

7. Evaluate exactly. Here log means base 10. (a) $\log(0.25) + \log(400)$ (b) $\frac{\log_7(64)}{\log_7(16)}$	7. (a) 5 pts (b) 5 pts
8. Solve the nonlinear system. $y = x^2$ $y = 4x + 32$	8. 10 pts
9. Find all zeros. $p(x) = x^3 - 2x^2 - 4x + 3$	9. 10 pts
10. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_2(x - 3) + 2$	10. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts

74618

TEST PAGE 1

1. (a)
$[-4, \infty)$ 2 pts

(b)
$[-6, \infty)$ 2 pts

(c)
$(5, 0)$ 1 pts

(d)
$(0, -2)$ 1 pts

(e)
$x \to \infty : f(x) \to \infty$ 3 pts

(f)
$x \to -4^+ : f(x) \to -6$ 1 pts

2.
C

10 pts

3.
$\frac{1}{2}(x+1)(x-3)^2$

10 pts

TEST PAGE 2

4. (a)
$\mathbb{R} \setminus \{-3, 6\}$ 2 pts

(b)
$(-3, -\frac{1}{9})$ 2 pts

(c)
$x = 6$ 1 pts

(d)
$y = 1$ 1 pts

(e)
$(-4, 0)$ 2 pts

(f)
$(0, -\frac{2}{3})$ 2 pts

5. (a)
C

3 pts

(b)
A

2 pts

(c)
D

3 pts

(d)
B

2 pts

6.
A

10 pts

TEST PAGE 3

7. (a)
2 5 pts

(b)
$\frac{3}{2}$ 5 pts

8.
$(-4, 16); (8, 64)$

10 pts

9.
$x = 3, -\frac{1}{2} + \frac{\sqrt{5}}{2}, -\frac{\sqrt{5}}{2} - \frac{1}{2}$ 10 pts

10. (a)
$(3, \infty)$ 2 pts

(b)
$(-\infty, \infty)$ 2 pts

(c)
$x = 3$ 1 pts

(d)
$(\frac{13}{4}, 0)$ 1 pts

(e)
$x \to 3^+ : f(x) \to -\infty$ 2 pts

(f)
$x \to \infty : f(x) \to \infty$ 2 pts

MATH 101 **76018**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

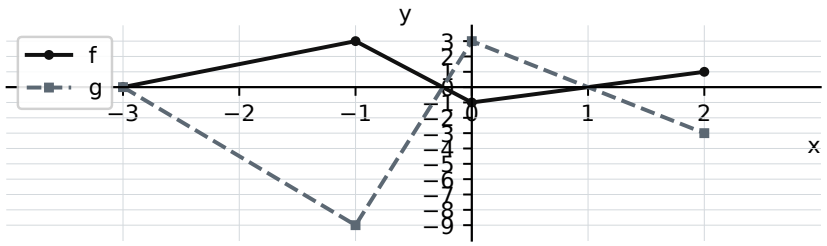
1. The solid graph is f and the dashed graph is g. Select the expression for g(x).

$A.\quad g(x) = 3f(x)$

$B.\quad g(x) = f(-3x)$

$C.\quad g(x) = -3f(x)$

$D.\quad g(x) = f(x) - 3$



1.

10 pts

2. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept.

$$f(x) = \frac{(x+3)(x+4)}{(x-6)(x+3)}$$

2. (a)

2 pts

(b)

2 pts

(c)

1 pts

(d)

1 pts

(e)

2 pts

(f)

2 pts

Test continues on the other side of the sheet.

MATH 101 **76018**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

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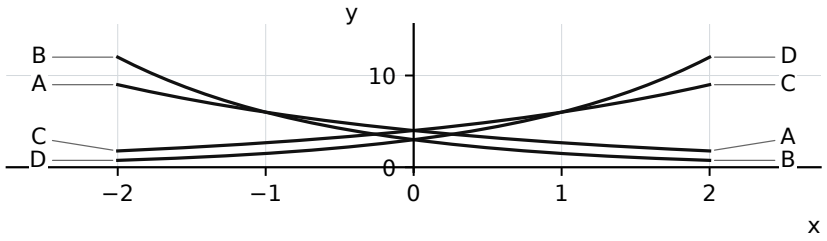
3. Match each function to graph A, B, C, or D. Each letter is used once.

(a) $4(\frac{3}{2})^x$

(b) $4(\frac{2}{3})^x$

(c) $3(2)^x$

(d) $3(\frac{1}{2})^x$



3. (a)

3 pts

(b)

2 pts

(c)

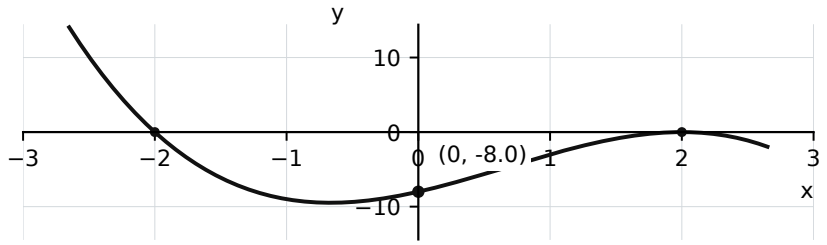
3 pts

(d)

2 pts

---

4. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form.



4.

10 pts

---

5. Evaluate exactly. Here log means base 10.

(a) $\log(0.8) + \log(1250)$

(b) $\frac{\log_7(64)}{\log_7(16)}$

5. (a)

5 pts

(b)

5 pts

Test continues on the next sheet.

MATH 101 **76018**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

6. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 2\sqrt{x+9} - 6$	6. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts
7. Solve exactly. $\sqrt{(x-2)^2 + 20} = 6$	7. 10 pts
8. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_3(x-3) + 2$	8. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts

Test continues on the other side of the sheet.

MATH 101 **76018**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

9. Which pair satisfies h(x) = f(g(x))? $h(x) = (4x + 4)^2 + 5$ $A.\quad f(x) = x^2 + 5,\ g(x) = 4x + 4$ $B.\quad f(x) = 4x + 4,\ g(x) = x^2 + 5$ $C.\quad f(x) = x^2 + 4,\ g(x) = 4x + 5$ $D.\quad f(x) = 4x^2 + 5,\ g(x) = x + 4$	9. 10 pts
10. Find all zeros. $p(x) = x^3 + 5x^2 + 7x + 2$	10. 10 pts

76018

TEST PAGE 1

1.

C

10 pts

2. (a)

$\mathbb{R} \setminus \{-3, 6\}$ 2 pts

(b)

$(-3, -\frac{1}{9})$ 2 pts

(c)

$x = 6$ 1 pts

(d)

$y = 1$ 1 pts

(e)

$(-4, 0)$ 2 pts

(f)

$(0, -\frac{2}{3})$ 2 pts

TEST PAGE 2

3. (a)

C

3 pts

(b)

A

2 pts

(c)

D

3 pts

(d)

B

2 pts

4.

$-1(x+2)(x-2)^2$

10 pts

5. (a)

3 5 pts

(b)

$\frac{3}{2}$ 5 pts

TEST PAGE 3

6. (a)

$[-9, \infty)$ 2 pts

(b)

$[-6, \infty)$ 2 pts

(c)

$(0, 0)$ 1 pts

(d)

$(0, 0)$ 1 pts

(e)

$x \to \infty : f(x) \to \infty$ 3 pts

(f)

$x \to -9^+ : f(x) \to -6$ 1 pts

7.

$x = -2, 6$

10 pts

8. (a)

$(3, \infty)$ 2 pts

(b)

$(-\infty, \infty)$ 2 pts

(c)

$x = 3$ 1 pts

(d)

$(\frac{28}{9}, 0)$ 1 pts

(e)

$x \to 3^+ : f(x) \to -\infty$ 2 pts

(f)

$x \to \infty : f(x) \to \infty$ 2 pts

76018

TEST PAGE 4

9.

A

10 pts

10.

$x = -2,\ -\frac{3}{2} - \frac{\sqrt{5}}{2},\ -\frac{3}{2} + \frac{\sqrt{5}}{2}$

10 pts

MATH 101 **20580**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

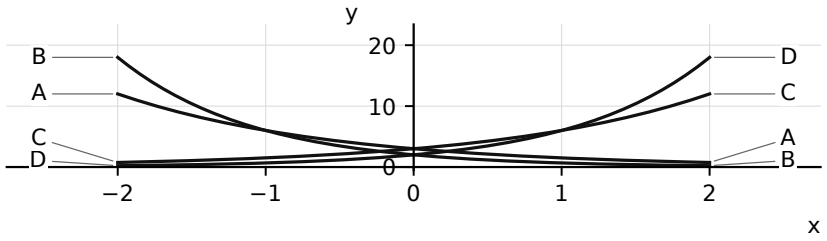
1. Match each function to graph A, B, C, or D. Each letter is used once.

(a) $3(2)^x$

(b) $3(\frac{1}{2})^x$

(c) $2(3)^x$

(d) $2(\frac{1}{3})^x$



1. (a)

3 pts

(b)

2 pts

(c)

3 pts

(d)

2 pts

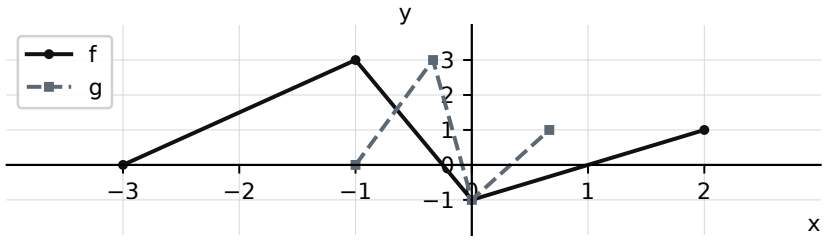
2. The solid graph is f and the dashed graph is g. Select the expression for g(x).

$A.$ $g(x) = f(x/3)$

$B.$ $g(x) = 3f(x)$

$C.$ $g(x) = f(3x)$

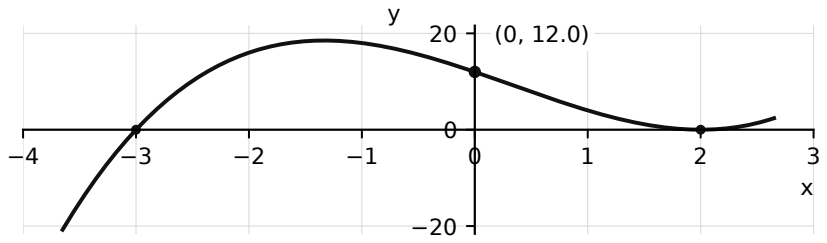
$D.$ $g(x) = f(x - 3)$



2.

10 pts

3. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form.



3.

10 pts

Test continues on the other side of the sheet.

MATH 101 **20580**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Find all zeros. $p(x) = x^3 + 3x^2 - 2$	4. 10 pts
5. Which pair satisfies h(x) = f(g(x))? $h(x) = (4x + 4)^2 + 5$ $A.\quad f(x) = x^2 + 5,\ g(x) = 4x + 4$ $B.\quad f(x) = 4x + 4,\ g(x) = x^2 + 5$ $C.\quad f(x) = x^2 + 4,\ g(x) = 4x + 5$ $D.\quad f(x) = 4x^2 + 5,\ g(x) = x + 4$	5. 10 pts
6. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 2\sqrt{x + 16} - 6$	6. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts

Test continues on the next sheet.

MATH 101 **20580**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

7. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_4(x-3) + 2$	7. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
8. Evaluate exactly. Here log means base 10. $(a)\quad \log(0.05) + \log(200000)$ $(b)\quad \frac{\log_7(64)}{\log_7(16)}$	8. (a) 5 pts (b) 5 pts
9. Solve exactly. $8^{x-3} = 4^{x+5}$	9. 10 pts
10. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x+3)(x+4)}{(x-6)(x+3)}$	10. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts

20580

TEST PAGE 1

1. (a)

C

3 pts

(b)

A

2 pts

(c)

D

3 pts

(d)

B

2 pts

2.

C

10 pts

3.

$1(x+3)(x-2)^2$

10 pts

TEST PAGE 2

4.

$x=-1,\ -1+\sqrt{3},\ -\sqrt{3}-1$

10 pts

5.

A

10 pts

6. (a)

$[-16,\infty)$ 2 pts

(b)

$[-6,\infty)$ 2 pts

(c)

$(-7,0)$ 1 pts

(d)

$(0,2)$ 1 pts

(e)

$x\to\infty : f(x)\to\infty$ 3 pts

(f)

$x\to -16^{+} : f(x)\to -6$ 1 pts

TEST PAGE 3

7. (a)

$(3,\infty)$ 2 pts

(b)

$(-\infty,\infty)$ 2 pts

(c)

$x=3$ 1 pts

(d)

$(\frac{49}{16},0)$ 1 pts

(e)

$x\to 3^{+} : f(x)\to -\infty$ 2 pts

(f)

$x\to\infty : f(x)\to\infty$ 2 pts

8. (a)

4 5 pts

(b)

$\frac{3}{2}$ 5 pts

9.

$x=19$

10 pts

10. (a)

$\mathbb{R}\setminus\{-3,6\}$ 2 pts

(b)

$(-3,\ -\frac{1}{9})$ 2 pts

(c)

$x=6$ 1 pts

(d)

$y=1$ 1 pts

(e)

$(-4,0)$ 2 pts

(f)

$(0,\ -\frac{2}{3})$ 2 pts

MATH 101 **76407**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

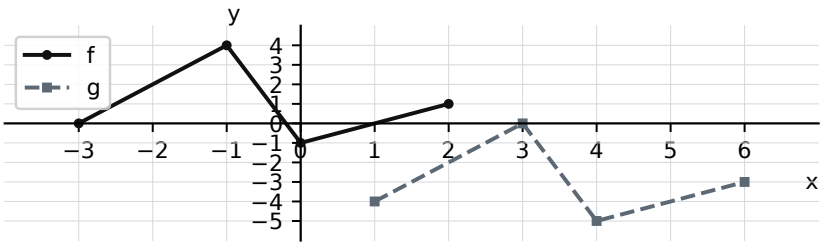
1. The solid graph is f and the dashed graph is g. Select the expression for g(x).

$A.\quad g(x) = f(x-4) - 4$

$B.\quad g(x) = f(x+4) + 4$

$C.\quad g(x) = f(x-4) + 4$

$D.\quad g(x) = f(x+4) - 4$



1.

10 pts

2. Evaluate exactly. Here log means base 10.

$(a)\quad \log(0.25) + \log(400)$

$(b)\quad \frac{\log_8(125)}{\log_8(25)}$

2. (a)

5 pts

(b)

5 pts

3. Find all zeros.

$p(x) = x^3 - 2x^2 - 4x + 3$

3.

10 pts

4. Which pair satisfies h(x) = f(g(x))?

$h(x) = (5x+5)^2 + 6$

$A.\quad f(x) = 5x^2 + 6,\ g(x) = x + 5$

$B.\quad f(x) = 5x + 5,\ g(x) = x^2 + 6$

$C.\quad f(x) = x^2 + 5,\ g(x) = 5x + 6$

$D.\quad f(x) = x^2 + 6,\ g(x) = 5x + 5$

4.

10 pts

Test continues on the other side of the sheet.

MATH 101 **76407**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

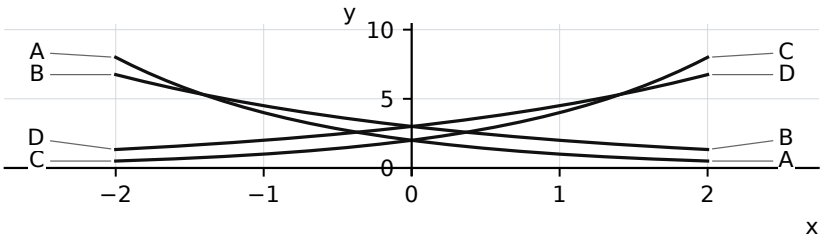
5. Match each function to graph A, B, C, or D. Each letter is used once.

(a) $2(2)^x$

(b) $2(\frac{1}{2})^x$

(c) $3(\frac{3}{2})^x$

(d) $3(\frac{2}{3})^x$



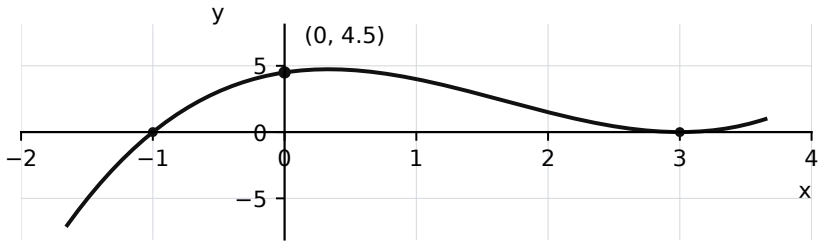
5. (a) 3 pts

(b) 2 pts

(c) 3 pts

(d) 2 pts

6. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form.



6. 10 pts

7. Solve exactly.

$$\log_4(x-3) - \log_4(x-6) = 1$$

7. 10 pts

Test continues on the next sheet.

MATH 101 **76407**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

8. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_2(x-4)+3$	8. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
9. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x+4)(x+5)}{(x-7)(x+4)}$	9. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts

Test continues on the other side of the sheet.

MATH 101 **76407**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

10. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 3\sqrt{x+4} - 12$	10. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts

76407

TEST PAGE 1

1.

A

10 pts

2. (a)

2

5 pts

(b)

$\frac{3}{2}$

5 pts

3.

$x = 3, -\frac{1}{2} + \frac{\sqrt{5}}{2}, -\frac{\sqrt{5}}{2} - \frac{1}{2}$

10 pts

4.

D

10 pts

TEST PAGE 2

5. (a)

C

3 pts

(b)

A

2 pts

(c)

D

3 pts

(d)

B

2 pts

6.

$\frac{1}{2}(x+1)(x-3)^2$

10 pts

7.

$x = 7$

10 pts

TEST PAGE 3

8. (a)

$(4, \infty)$

2 pts

(b)

$(-\infty, \infty)$

2 pts

(c)

$x = 4$

1 pts

(d)

$(\frac{33}{8}, 0)$

1 pts

(e)

$x \to 4^+ : f(x) \to -\infty$

2 pts

(f)

$x \to \infty : f(x) \to \infty$

2 pts

9. (a)

$\mathbb{R} \setminus \{-4, 7\}$

2 pts

(b)

$(-4, -\frac{1}{11})$

2 pts

(c)

$x = 7$

1 pts

(d)

$y = 1$

1 pts

(e)

$(-5, 0)$

2 pts

(f)

$(0, -\frac{5}{7})$

2 pts

76407

TEST PAGE 4

10. (a)
$[-4, \infty)$ 2 pts

(b)
$[-12, \infty)$ 2 pts

(c)
$(12, 0)$ 1 pts

(d)
$(0, -6)$ 1 pts

(e)
$x \to \infty : f(x) \to \infty$ 3 pts

(f)
$x \to -4^{+} : f(x) \to -12$ 1 pts

MATH 101 **74695**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Which pair satisfies h(x) = f(g(x))?

$h(x) = (5x + 5)^2 + 6$

$A.\quad f(x) = 5x^2 + 6,\ g(x) = x + 5$

$B.\quad f(x) = 5x + 5,\ g(x) = x^2 + 6$

$C.\quad f(x) = x^2 + 5,\ g(x) = 5x + 6$

$D.\quad f(x) = x^2 + 6,\ g(x) = 5x + 5$

1.

10 pts

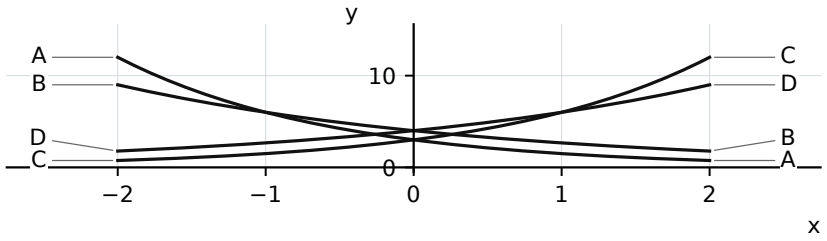
2. Match each function to graph A, B, C, or D. Each letter is used once.

$(a)\quad 3(2)^x$

$(b)\quad 3(\frac{1}{2})^x$

$(c)\quad 4(\frac{3}{2})^x$

$(d)\quad 4(\frac{2}{3})^x$



2. (a)

3 pts

(b)

2 pts

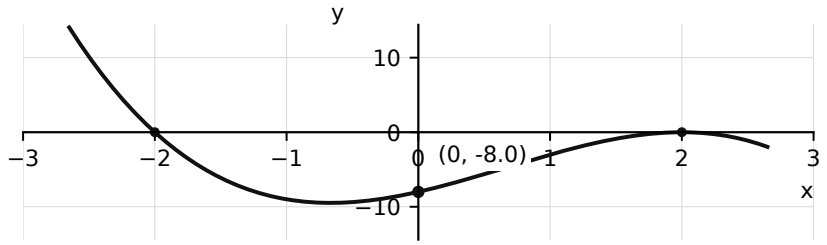
(c)

3 pts

(d)

2 pts

3. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form.



3.

10 pts

4. Evaluate exactly. Here log means base 10.

$(a)\quad \log(0.8) + \log(1250)$

$(b)\quad \frac{\log_8(125)}{\log_8(25)}$

4. (a)

5 pts

(b)

5 pts

Test continues on the other side of the sheet.

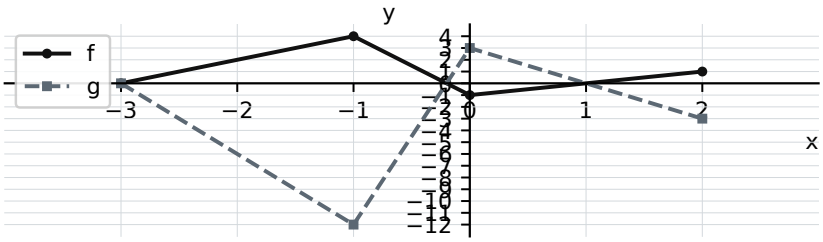
MATH 101 **74695**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

5. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x+4)(x+5)}{(x-7)(x+4)}$	5. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
6. The solid graph is f and the dashed graph is g. Select the expression for g(x). $A.\quad g(x) = -3f(x)$ $B.\quad g(x) = f(x) - 3$ $C.\quad g(x) = 3f(x)$ $D.\quad g(x) = f(-3x)$ 	6. 10 pts
7. Find all zeros. $p(x) = x^3 + 5x^2 + 7x + 2$	7. 10 pts

Test continues on the next sheet.

MATH 101 **74695**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

8. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 3\sqrt{x+9} - 12$	8. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts
9. Solve the nonlinear system. $x^2 + y^2 = 17$ $y = x + 3$	9. 10 pts
10. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_3(x-4) + 3$	10. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts

74695

TEST PAGE 1

1.

D

10 pts

2. (a)

C

3 pts

(b)

A

2 pts

(c)

D

3 pts

(d)

B

2 pts

3.

$-1(x+2)(x-2)^2$

10 pts

4. (a)

3

5 pts

(b)

$\frac{3}{2}$

5 pts

TEST PAGE 2

5. (a)

$\mathbb{R} \setminus \{-4, 7\}$

2 pts

(b)

$(-4, -\frac{1}{11})$

2 pts

(c)

$x = 7$

1 pts

(d)

$y = 1$

1 pts

(e)

$(-5, 0)$

2 pts

(f)

$(0, -\frac{5}{7})$

2 pts

6.

A

10 pts

7.

$x = -2, -\frac{3}{2} - \frac{\sqrt{5}}{2}, -\frac{3}{2} + \frac{\sqrt{5}}{2}$

10 pts

TEST PAGE 3

8. (a)

$[-9, \infty)$

2 pts

(b)

$[-12, \infty)$

2 pts

(c)

$(7, 0)$

1 pts

(d)

$(0, -3)$

1 pts

(e)

$x \to \infty : f(x) \to \infty$

3 pts

(f)

$x \to -9^+ : f(x) \to -12$

1 pts

9.

$(1, 4); (-4, -1)$

10 pts

10. (a)

$(4, \infty)$

2 pts

(b)

$(-\infty, \infty)$

2 pts

(c)

$x = 4$

1 pts

(d)

$(\frac{109}{27}, 0)$

1 pts

(e)

$x \to 4^+ : f(x) \to -\infty$

2 pts

(f)

$x \to \infty : f(x) \to \infty$

2 pts

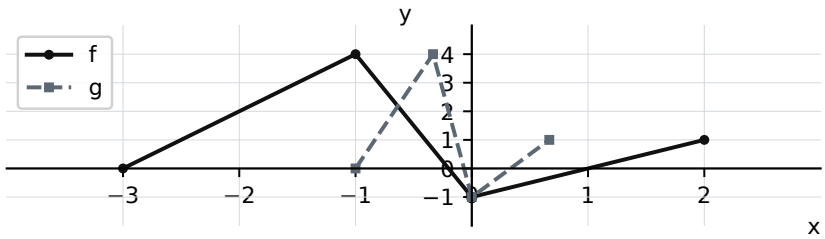
MATH 101 **48624**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Solve exactly. $\sqrt{x+2} = x-2$	1. 10 pts
2. The solid graph is f and the dashed graph is g. Select the expression for g(x). $A.\quad g(x) = f(3x)$ $B.\quad g(x) = f(x-3)$ $C.\quad g(x) = f(x/3)$ $D.\quad g(x) = 3f(x)$ 	2. 10 pts
3. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x+4)(x+5)}{(x-7)(x+4)}$	3. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts

Test continues on the other side of the sheet.

MATH 101 **48624**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_4(x-4) + 3$	4. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
5. Find all zeros. $p(x) = x^3 + 3x^2 - 2$	5. 10 pts
6. Evaluate exactly. Here log means base 10. (a) $\log(0.05) + \log(200000)$ (b) $\frac{\log_8(125)}{\log_8(25)}$	6. (a) 5 pts (b) 5 pts
7. Which pair satisfies h(x) = f(g(x))? $h(x) = (5x+5)^2 + 6$ $A.\quad f(x) = 5x^2 + 6,\ g(x) = x + 5$ $B.\quad f(x) = 5x + 5,\ g(x) = x^2 + 6$ $C.\quad f(x) = x^2 + 5,\ g(x) = 5x + 6$ $D.\quad f(x) = x^2 + 6,\ g(x) = 5x + 5$	7. 10 pts

Test continues on the next sheet.

MATH 101 **48624**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

8. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end.

$f(x) = 3\sqrt{x+16} - 12$

8. (a) 2 pts

(b) 2 pts

(c) 1 pts

(d) 1 pts

(e) 3 pts

(f) 1 pts

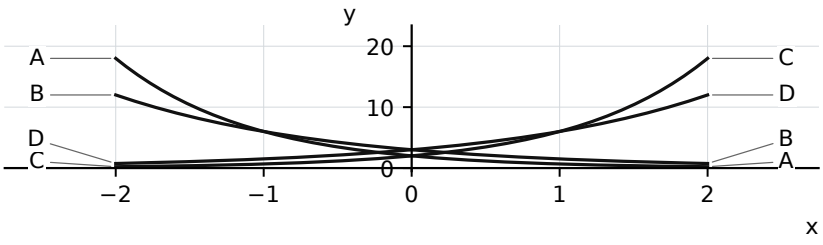
9. Match each function to graph A, B, C, or D. Each letter is used once.

$(a)\quad 2(3)^x$

$(b)\quad 2(\frac{1}{3})^x$

$(c)\quad 3(2)^x$

$(d)\quad 3(\frac{1}{2})^x$



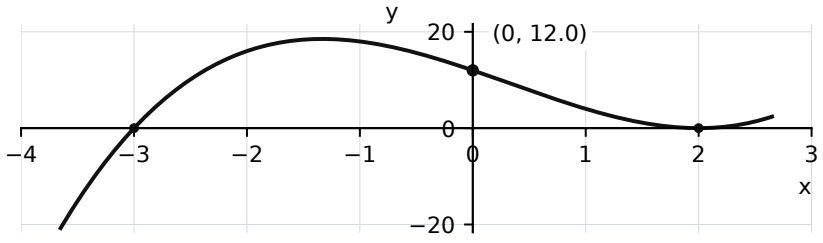
9. (a) 3 pts

(b) 2 pts

(c) 3 pts

(d) 2 pts

10. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form.



10. 10 pts

48624

TEST PAGE 1

1.

$x = \frac{\sqrt{17}}{2} + \frac{5}{2}$

10 pts

2.

A

10 pts

3. (a)

$\mathbb{R} \setminus \{-4, 7\}$ 2 pts

(b)

$(-4, -\frac{1}{11})$ 2 pts

(c)

$x = 7$ 1 pts

(d)

$y = 1$ 1 pts

(e)

$(-5, 0)$ 2 pts

(f)

$(0, -\frac{5}{7})$ 2 pts

TEST PAGE 2

4. (a)

$(4, \infty)$ 2 pts

(b)

$(-\infty, \infty)$ 2 pts

(c)

$x = 4$ 1 pts

(d)

$(\frac{257}{64}, 0)$ 1 pts

(e)

$x \to 4^+ : f(x) \to -\infty$ 2 pts

(f)

$x \to \infty : f(x) \to \infty$ 2 pts

5.

$x = -1, -1 + \sqrt{3}, -\sqrt{3} - 1$

10 pts

6. (a)

4 5 pts

(b)

$\frac{3}{2}$ 5 pts

7.

D

10 pts

TEST PAGE 3

8. (a)

$[-16, \infty)$ 2 pts

(b)

$[-12, \infty)$ 2 pts

(c)

$(0, 0)$ 1 pts

(d)

$(0, 0)$ 1 pts

(e)

$x \to \infty : f(x) \to \infty$ 3 pts

(f)

$x \to -16^+ : f(x) \to -12$ 1 pts

9. (a)

C

3 pts

(b)

A

2 pts

(c)

D

3 pts

(d)

B

2 pts

10.

$1(x + 3)(x - 2)^2$

10 pts

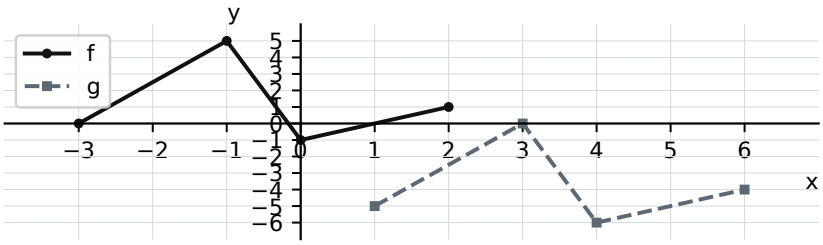
MATH 101 **11162**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Evaluate exactly. Here log means base 10. $(a)\quad \log(0.25)+\log(400)$ $(b)\quad \frac{\log_9(216)}{\log_9(36)}$	1. (a) 5 pts (b) 5 pts
2. Which pair satisfies h(x) = f(g(x))? $h(x)=(6x+6)^2+7$ $A.\quad f(x)=6x^2+7,\ g(x)=x+6$ $B.\quad f(x)=x^2+7,\ g(x)=6x+6$ $C.\quad f(x)=x^2+6,\ g(x)=6x+7$ $D.\quad f(x)=6x+6,\ g(x)=x^2+7$	2. 10 pts
3. Find all zeros. $p(x)=x^3-2x^2-4x+3$	3. 10 pts
4. The solid graph is f and the dashed graph is g. Select the expression for g(x). $A.\quad g(x)=f(x-4)+5$ $B.\quad g(x)=f(x+4)-5$ $C.\quad g(x)=f(x+4)+5$ $D.\quad g(x)=f(x-4)-5$ 	4. 10 pts

Test continues on the other side of the sheet.

MATH 101 **11162**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

5. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 4\sqrt{x+4} - 20$	5. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts
6. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x+5)(x+6)}{(x-8)(x+5)}$	6. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts

Test continues on the next sheet.

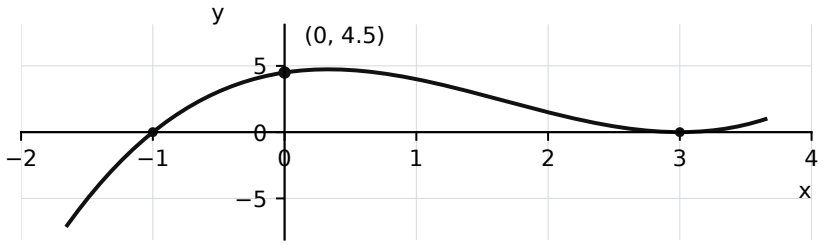
MATH 101 **11162**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

7. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_2(x-5)+1$	7. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
8. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form. 	8. 10 pts

Test continues on the other side of the sheet.

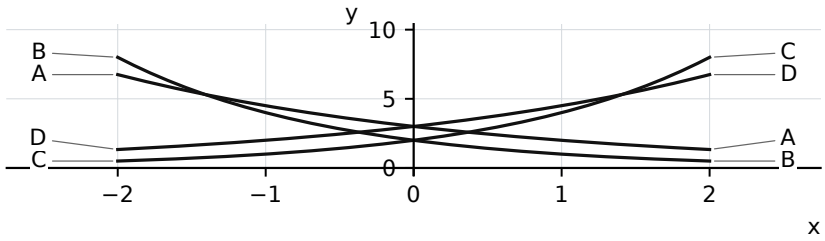
MATH 101 **11162**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

9. Match each function to graph A, B, C, or D. Each letter is used once. $(a)\quad 2(2)^x$ $(b)\quad 3(\frac{2}{3})^x$ $(c)\quad 3(\frac{3}{2})^x$ $(d)\quad 2(\frac{1}{2})^x$ 	9. (a) 3 pts (b) 2 pts (c) 3 pts (d) 2 pts
10. Solve exactly. $8^{x-2} = 4^{x+3}$	10. 10 pts

11162

TEST PAGE 1

1. (a)

2 — 5 pts

(b)

$\frac{3}{2}$ — 5 pts

2.

B

10 pts

3.

$x = 3,\ -\frac{1}{2} + \frac{\sqrt{5}}{2},\ -\frac{\sqrt{5}}{2} - \frac{1}{2}$ — 10 pts

4.

D

10 pts

TEST PAGE 2

5. (a)

$[-4, \infty)$ — 2 pts

(b)

$[-20, \infty)$ — 2 pts

(c)

$(21, 0)$ — 1 pts

(d)

$(0, -12)$ — 1 pts

(e)

$x \to \infty : f(x) \to \infty$ — 3 pts

(f)

$x \to -4^+ : f(x) \to -20$ — 1 pts

6. (a)

$\mathbb{R} \setminus \{-5, 8\}$ — 2 pts

(b)

$(-5, -\frac{1}{13})$ — 2 pts

(c)

$x = 8$ — 1 pts

(d)

$y = 1$ — 1 pts

(e)

$(-6, 0)$ — 2 pts

(f)

$(0, -\frac{3}{4})$ — 2 pts

TEST PAGE 3

7. (a)

$(5, \infty)$ — 2 pts

(b)

$(-\infty, \infty)$ — 2 pts

(c)

$x = 5$ — 1 pts

(d)

$(\frac{11}{2}, 0)$ — 1 pts

(e)

$x \to 5^+ : f(x) \to -\infty$ — 2 pts

(f)

$x \to \infty : f(x) \to \infty$ — 2 pts

8.

$\frac{1}{2}(x+1)(x-3)^2$

10 pts

11162

TEST PAGE 4

9. (a)

C

3 pts

(b)

A

2 pts

(c)

D

3 pts

(d)

B

2 pts

10.

$x = 12$

10 pts

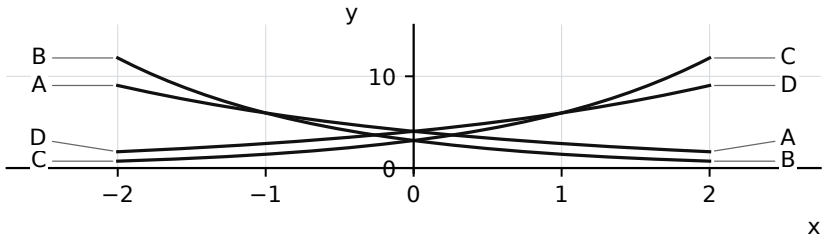
MATH 101 **54236**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find all zeros. $p(x) = x^3 + 5x^2 + 7x + 2$	1. 10 pts
2. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x+5)(x+6)}{(x-8)(x+5)}$	2. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
3. Match each function to graph A, B, C, or D. Each letter is used once. $(a)\quad 3(2)^x$ $(b)\quad 4(\frac{2}{3})^x$ $(c)\quad 4(\frac{3}{2})^x$ $(d)\quad 3(\frac{1}{2})^x$ 	3. (a) 3 pts (b) 2 pts (c) 3 pts (d) 2 pts

Test continues on the other side of the sheet.

MATH 101 **54236**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Which pair satisfies h(x) = f(g(x))? $h(x) = (6x + 6)^2 + 7$ $A.\quad f(x) = 6x^2 + 7,\ g(x) = x + 6$ $B.\quad f(x) = x^2 + 7,\ g(x) = 6x + 6$ $C.\quad f(x) = x^2 + 6,\ g(x) = 6x + 7$ $D.\quad f(x) = 6x + 6,\ g(x) = x^2 + 7$	4. 10 pts
5. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_3(x - 5) + 1$	5. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
6. Solve exactly. $\log_2(x - 3) + \log_2(x - 6) = 2$	6. 10 pts

Test continues on the next sheet.

MATH 101 **54236**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

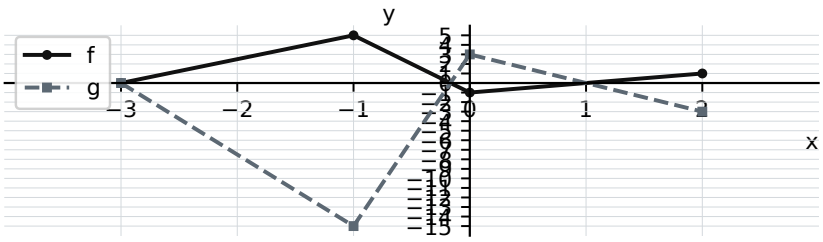
7. The solid graph is f and the dashed graph is g. Select the expression for g(x).

A. $g(x) = 3f(x)$

B. $g(x) = f(-3x)$

C. $g(x) = f(x) - 3$

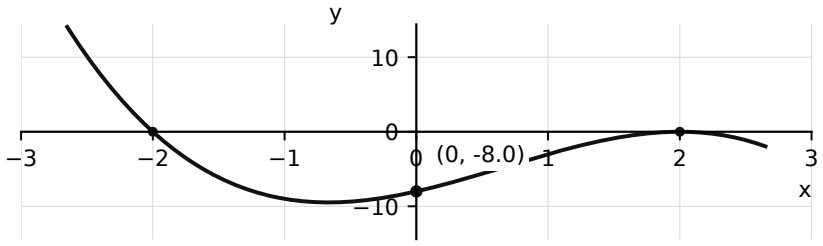
D. $g(x) = -3f(x)$



7.

10 pts

8. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form.



8.

10 pts

9. Evaluate exactly. Here log means base 10.

(*a*) $\log(0.8) + \log(1250)$

(*b*) $\frac{\log_9(216)}{\log_9(36)}$

9. (a)

5 pts

(b)

5 pts

Test continues on the other side of the sheet.

MATH 101 **54236**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

10. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 4\sqrt{x+9} - 20$	10. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts

54236

TEST PAGE 1

1.

$x = -2,\ -\frac{3}{2} - \frac{\sqrt{5}}{2},\ -\frac{3}{2} + \frac{\sqrt{5}}{2}$

10 pts

2. (a)

$\mathbb{R} \setminus \{-5, 8\}$

2 pts

(b)

$(-5,\ -\frac{1}{13})$

2 pts

(c)

$x = 8$

1 pts

(d)

$y = 1$

1 pts

(e)

$(-6, 0)$

2 pts

(f)

$(0,\ -\frac{3}{4})$

2 pts

3. (a)

C

3 pts

(b)

A

2 pts

(c)

D

3 pts

(d)

B

2 pts

TEST PAGE 2

4.

B

10 pts

5. (a)

$(5, \infty)$

2 pts

(b)

$(-\infty, \infty)$

2 pts

(c)

$x = 5$

1 pts

(d)

$(\frac{16}{3}, 0)$

1 pts

(e)

$x \to 5^+ : f(x) \to -\infty$

2 pts

(f)

$x \to \infty : f(x) \to \infty$

2 pts

6.

$x = 7$

10 pts

TEST PAGE 3

7.

D

10 pts

8.

$-1(x+2)(x-2)^2$

10 pts

9. (a)

3

5 pts

(b)

$\frac{3}{2}$

5 pts

54236

TEST PAGE 4

10. (a)
$[-9, \infty)$ 2 pts

(b)
$[-20, \infty)$ 2 pts

(c)
$(16, 0)$ 1 pts

(d)
$(0, -8)$ 1 pts

(e)
$x \to \infty : f(x) \to \infty$ 3 pts

(f)
$x \to -9^{+} : f(x) \to -20$ 1 pts

MATH 101 **97643**

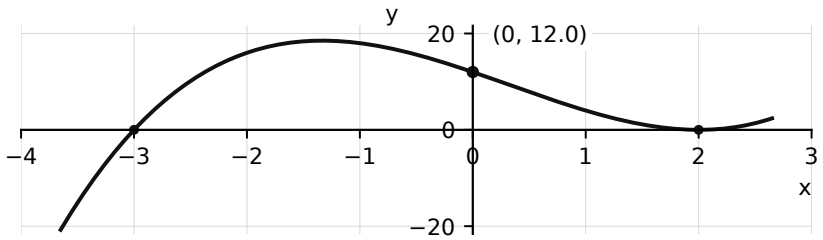
Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form.



1.

10 pts

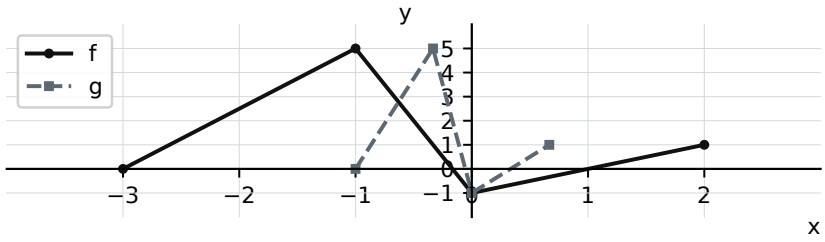
2. The solid graph is f and the dashed graph is g. Select the expression for g(x).

$A.\quad g(x) = f(x/3)$

$B.\quad g(x) = 3f(x)$

$C.\quad g(x) = f(x-3)$

$D.\quad g(x) = f(3x)$



2.

10 pts

3. Find all zeros.

$p(x) = x^3 + 3x^2 - 2$

3.

10 pts

4. Evaluate exactly. Here log means base 10.

$(a)\quad \log(0.05) + \log(200000)$

$(b)\quad \dfrac{\log_9(216)}{\log_9(36)}$

4. (a)

5 pts

(b)

5 pts

Test continues on the other side of the sheet.

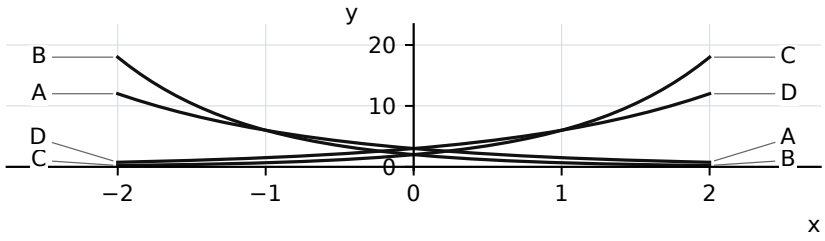
MATH 101 **97643**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

5. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x+5)(x+6)}{(x-8)(x+5)}$	5. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
6. Solve the nonlinear system. $y = x^2$ $y = 3x + 18$	6. 10 pts
7. Match each function to graph A, B, C, or D. Each letter is used once. $(a)\quad 2(3)^x$ $(b)\quad 3(\frac{1}{2})^x$ $(c)\quad 3(2)^x$ $(d)\quad 2(\frac{1}{3})^x$ 	7. (a) 3 pts (b) 2 pts (c) 3 pts (d) 2 pts

Test continues on the next sheet.

MATH 101 **97643**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

8. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 4\sqrt{x+16} - 20$	8. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts
9. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_4(x-5) + 1$	9. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
10. Which pair satisfies h(x) = f(g(x))? $h(x) = (6x+6)^2 + 7$ $A.\quad f(x) = 6x^2 + 7,\ g(x) = x + 6$ $B.\quad f(x) = x^2 + 7,\ g(x) = 6x + 6$ $C.\quad f(x) = x^2 + 6,\ g(x) = 6x + 7$ $D.\quad f(x) = 6x + 6,\ g(x) = x^2 + 7$	10. 10 pts

97643

TEST PAGE 1

1.

$1(x+3)(x-2)^2$

10 pts

2.

D

10 pts

3.

$x=-1,\ -1+\sqrt{3},\ -\sqrt{3}-1$

10 pts

4. (a)

4 — 5 pts

(b)

$\frac{3}{2}$ — 5 pts

TEST PAGE 2

5. (a)

$\mathbb{R} \setminus \{-5, 8\}$ — 2 pts

(b)

$(-5,\ -\frac{1}{13})$ — 2 pts

(c)

$x=8$ — 1 pts

(d)

$y=1$ — 1 pts

(e)

$(-6, 0)$ — 2 pts

(f)

$(0,\ -\frac{3}{4})$ — 2 pts

6.

$(-3, 9); (6, 36)$

10 pts

7. (a)

C

3 pts

(b)

A

2 pts

(c)

D

3 pts

(d)

B

2 pts

TEST PAGE 3

8. (a)

$[-16, \infty)$ — 2 pts

(b)

$[-20, \infty)$ — 2 pts

(c)

$(9, 0)$ — 1 pts

(d)

$(0,\ -4)$ — 1 pts

(e)

$x \to \infty : f(x) \to \infty$ — 3 pts

(f)

$x \to -16^+ : f(x) \to -20$ — 1 pts

9. (a)

$(5, \infty)$ — 2 pts

(b)

$(-\infty, \infty)$ — 2 pts

(c)

$x=5$ — 1 pts

(d)

$(\frac{21}{4}, 0)$ — 1 pts

(e)

$x \to 5^+ : f(x) \to -\infty$ — 2 pts

(f)

$x \to \infty : f(x) \to \infty$ — 2 pts

10.

B

10 pts

MATH 101 **64047**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

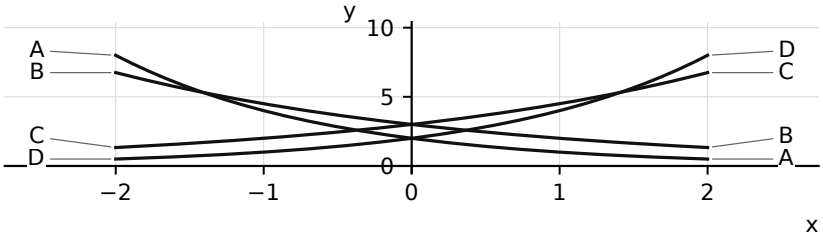
1. Match each function to graph A, B, C, or D. Each letter is used once.

(*a*) $3(\frac{3}{2})^x$

(*b*) $2(\frac{1}{2})^x$

(*c*) $2(2)^x$

(*d*) $3(\frac{2}{3})^x$



1. (a) 3 pts

(b) 2 pts

(c) 3 pts

(d) 2 pts

2. Evaluate exactly. Here log means base 10.

(*a*) $\log(0.25) + \log(400)$

(*b*) $\frac{\log_{10}(343)}{\log_{10}(49)}$

2. (a) 5 pts

(b) 5 pts

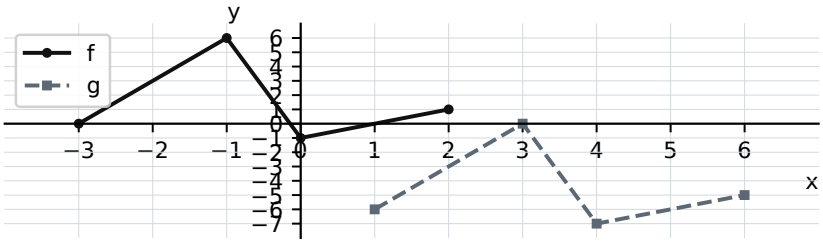
3. The solid graph is f and the dashed graph is g. Select the expression for g(x).

A. $g(x) = f(x-4) - 6$

B. $g(x) = f(x+4) + 6$

C. $g(x) = f(x-4) + 6$

D. $g(x) = f(x+4) - 6$



3. 10 pts

Test continues on the other side of the sheet.

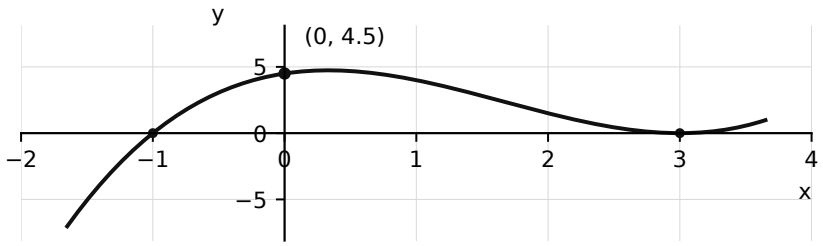
MATH 101 **64047**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form. 	4. 10 pts
5. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_2(x - 6) + 2$	5. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
6. Find all zeros. $p(x) = x^3 - 2x^2 - 4x + 3$	6. 10 pts
7. Solve exactly. $\sqrt{(x-3)^2 + 9} = 5$	7. 10 pts

Test continues on the next sheet.

MATH 101 **64047**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

8. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 5\sqrt{x+4} - 30$	8. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts
9. Which pair satisfies h(x) = f(g(x))? $h(x) = (7x+7)^2 + 8$ $A.\quad f(x) = 7x+7,\ g(x) = x^2+8$ $B.\quad f(x) = 7x^2+8,\ g(x) = x+7$ $C.\quad f(x) = x^2+8,\ g(x) = 7x+7$ $D.\quad f(x) = x^2+7,\ g(x) = 7x+8$	9. 10 pts
10. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x+6)(x+7)}{(x-9)(x+6)}$	10. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts

64047

TEST PAGE 1

1. (a)

C

3 pts

(b)

A

2 pts

(c)

D

3 pts

(d)

B

2 pts

2. (a)

2

5 pts

(b)

$\frac{3}{2}$

5 pts

3.

A

10 pts

TEST PAGE 2

4.

$\frac{1}{2}(x+1)(x-3)^2$

10 pts

5. (a)

$(6, \infty)$

2 pts

(b)

$(-\infty, \infty)$

2 pts

(c)

$x = 6$

1 pts

(d)

$(\frac{25}{4}, 0)$

1 pts

(e)

$x \to 6^+ : f(x) \to -\infty$

2 pts

(f)

$x \to \infty : f(x) \to \infty$

2 pts

6.

$x = 3, -\frac{1}{2} + \frac{\sqrt{5}}{2}, -\frac{\sqrt{5}}{2} - \frac{1}{2}$

10 pts

7.

$x = -1, 7$

10 pts

TEST PAGE 3

8. (a)

$[-4, \infty)$

2 pts

(b)

$[-30, \infty)$

2 pts

(c)

$(32, 0)$

1 pts

(d)

$(0, -20)$

1 pts

(e)

$x \to \infty : f(x) \to \infty$

3 pts

(f)

$x \to -4^+ : f(x) \to -30$

1 pts

9.

C

10 pts

10. (a)

$\mathbb{R} \setminus \{-6, 9\}$

2 pts

(b)

$(-6, -\frac{1}{15})$

2 pts

(c)

$x = 9$

1 pts

(d)

$y = 1$

1 pts

(e)

$(-7, 0)$

2 pts

(f)

$(0, -\frac{7}{9})$

2 pts

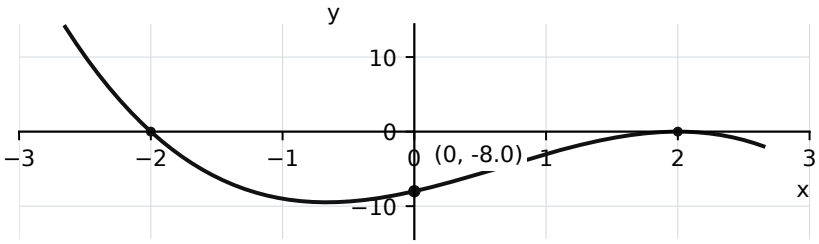
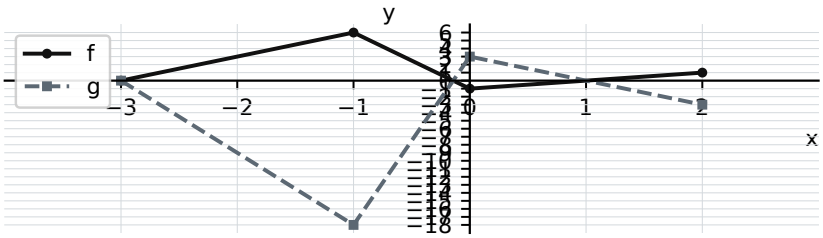
MATH 101 **84847**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find all zeros. $p(x) = x^3 + 5x^2 + 7x + 2$	1. 10 pts
2. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form. 	2. 10 pts
3. Evaluate exactly. Here log means base 10. $(a)\quad \log(0.8) + \log(1250)$ $(b)\quad \frac{\log_{10}(343)}{\log_{10}(49)}$	3. (a) 5 pts (b) 5 pts
4. The solid graph is f and the dashed graph is g. Select the expression for g(x). $A.\quad g(x) = -3f(x)$ $B.\quad g(x) = f(x) - 3$ $C.\quad g(x) = 3f(x)$ $D.\quad g(x) = f(-3x)$ 	4. 10 pts

Test continues on the other side of the sheet.

MATH 101 **84847**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

5. Which pair satisfies h(x) = f(g(x))?

$h(x) = (7x + 7)^2 + 8$

$A.\quad f(x) = 7x + 7,\ g(x) = x^2 + 8$

$B.\quad f(x) = 7x^2 + 8,\ g(x) = x + 7$

$C.\quad f(x) = x^2 + 8,\ g(x) = 7x + 7$

$D.\quad f(x) = x^2 + 7,\ g(x) = 7x + 8$

5.

10 pts

6. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end.

$f(x) = 5\sqrt{x + 9} - 30$

6. (a)

2 pts

(b)

2 pts

(c)

1 pts

(d)

1 pts

(e)

3 pts

(f)

1 pts

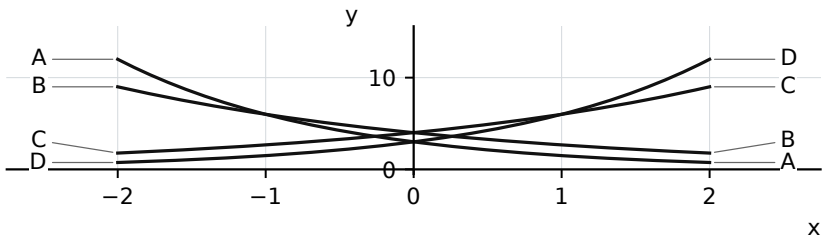
7. Match each function to graph A, B, C, or D. Each letter is used once.

$(a)\quad 4(\frac{3}{2})^x$

$(b)\quad 3(\frac{1}{2})^x$

$(c)\quad 3(2)^x$

$(d)\quad 4(\frac{2}{3})^x$



7. (a)

3 pts

(b)

2 pts

(c)

3 pts

(d)

2 pts

Test continues on the next sheet.

MATH 101 **84847**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

8. Solve exactly. $8^{x-3} = 4^{x+4}$	8. 10 pts
9. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x+6)(x+7)}{(x-9)(x+6)}$	9. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
10. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_3(x-6) + 2$	10. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts

84847

TEST PAGE 1

1.

$x = -2,\ -\frac{3}{2} - \frac{\sqrt{5}}{2},\ -\frac{3}{2} + \frac{\sqrt{5}}{2}$

10 pts

2.

$-1(x+2)(x-2)^2$

10 pts

3. (a)

3

5 pts

(b)

$\frac{3}{2}$

5 pts

4.

A

10 pts

TEST PAGE 2

5.

C

10 pts

6. (a)

$[-9, \infty)$

2 pts

(b)

$[-30, \infty)$

2 pts

(c)

$(27, 0)$

1 pts

(d)

$(0, -15)$

1 pts

(e)

$x \to \infty : f(x) \to \infty$

3 pts

(f)

$x \to -9^+ : f(x) \to -30$

1 pts

7. (a)

C

3 pts

(b)

A

2 pts

(c)

D

3 pts

(d)

B

2 pts

TEST PAGE 3

8.

$x = 17$

10 pts

9. (a)

$\mathbb{R} \setminus \{-6, 9\}$

2 pts

(b)

$(-6, -\frac{1}{15})$

2 pts

(c)

$x = 9$

1 pts

(d)

$y = 1$

1 pts

(e)

$(-7, 0)$

2 pts

(f)

$(0, -\frac{7}{9})$

2 pts

10. (a)

$(6, \infty)$

2 pts

(b)

$(-\infty, \infty)$

2 pts

(c)

$x = 6$

1 pts

(d)

$(\frac{55}{9}, 0)$

1 pts

(e)

$x \to 6^+ : f(x) \to -\infty$

2 pts

(f)

$x \to \infty : f(x) \to \infty$

2 pts

MATH 101 **59279**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept.

$$f(x) = \frac{(x+6)(x+7)}{(x-9)(x+6)}$$

1. (a) 2 pts

(b) 2 pts

(c) 1 pts

(d) 1 pts

(e) 2 pts

(f) 2 pts

2. Which pair satisfies h(x) = f(g(x))?

$$h(x) = (7x+7)^2 + 8$$

$A.\quad f(x) = 7x+7,\ g(x) = x^2+8$

$B.\quad f(x) = 7x^2+8,\ g(x) = x+7$

$C.\quad f(x) = x^2+8,\ g(x) = 7x+7$

$D.\quad f(x) = x^2+7,\ g(x) = 7x+8$

2. 10 pts

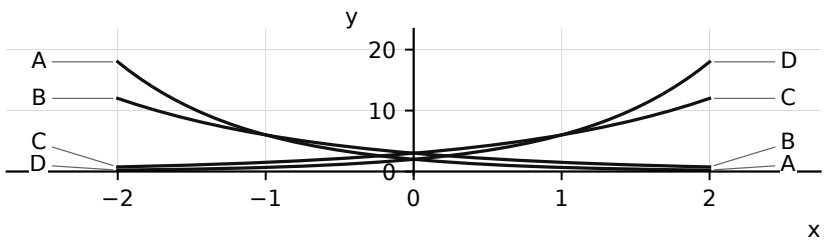
3. Match each function to graph A, B, C, or D. Each letter is used once.

$(a)\quad 3(2)^x$

$(b)\quad 2(\frac{1}{3})^x$

$(c)\quad 2(3)^x$

$(d)\quad 3(\frac{1}{2})^x$



3. (a) 3 pts

(b) 2 pts

(c) 3 pts

(d) 2 pts

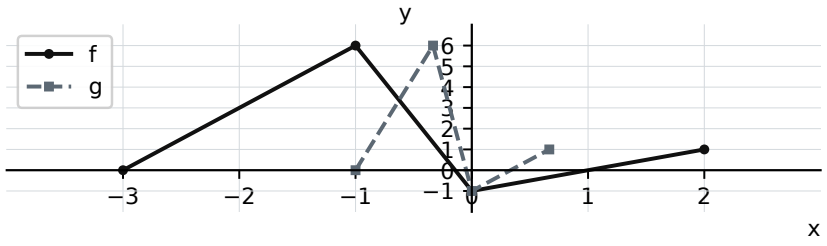
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Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_4(x - 6) + 2$	4. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
5. The solid graph is f and the dashed graph is g. Select the expression for g(x). *A*. $g(x) = f(3x)$ *B*. $g(x) = f(x - 3)$ *C*. $g(x) = f(x/3)$ *D*. $g(x) = 3f(x)$ 	5. 10 pts

Test continues on the next sheet.

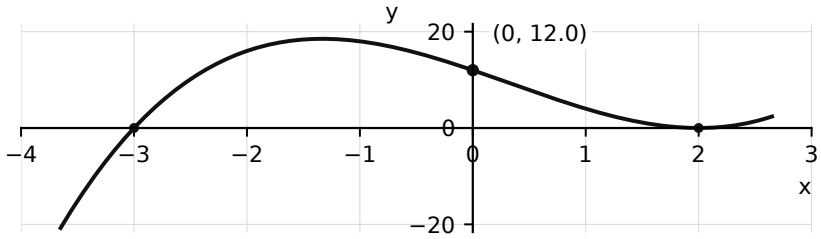
MATH 101 **59279**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

6. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 5\sqrt{x+16} - 30$	6. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts
7. Solve exactly. $\log_4(x-2) + \log_4(3) = 2$	7. 10 pts
8. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form. 	8. 10 pts
9. Evaluate exactly. Here log means base 10. $(a)\quad \log(0.05) + \log(200000)$ $(b)\quad \frac{\log_{10}(343)}{\log_{10}(49)}$	9. (a) 5 pts (b) 5 pts
10. Find all zeros. $p(x) = x^3 + 3x^2 - 2$	10. 10 pts

59279

TEST PAGE 1

1. (a)
$\mathbb{R} \setminus \{-6, 9\}$ 2 pts

(b)
$(-6, -\frac{1}{15})$ 2 pts

(c)
$x = 9$ 1 pts

(d)
$y = 1$ 1 pts

(e)
$(-7, 0)$ 2 pts

(f)
$(0, -\frac{7}{9})$ 2 pts

2.
C

10 pts

3. (a)
C

3 pts

(b)
A

2 pts

(c)
D

3 pts

(d)
B

2 pts

TEST PAGE 2

4. (a)
$(6, \infty)$ 2 pts

(b)
$(-\infty, \infty)$ 2 pts

(c)
$x = 6$ 1 pts

(d)
$(\frac{97}{16}, 0)$ 1 pts

(e)
$x \to 6^+ : f(x) \to -\infty$ 2 pts

(f)
$x \to \infty : f(x) \to \infty$ 2 pts

5.
A

10 pts

TEST PAGE 3

6. (a)
$[-16, \infty)$ 2 pts

(b)
$[-30, \infty)$ 2 pts

(c)
$(20, 0)$ 1 pts

(d)
$(0, -10)$ 1 pts

(e)
$x \to \infty : f(x) \to \infty$ 3 pts

(f)
$x \to -16^+ : f(x) \to -30$ 1 pts

7.
$x = \frac{22}{3}$

10 pts

8.
$1(x+3)(x-2)^2$

10 pts

9. (a)
4 5 pts

(b)
$\frac{3}{2}$ 5 pts

10.
$x = -1, -1+\sqrt{3}, -\sqrt{3}-1$

10 pts

MATH 101 **72324**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_2(x - 7) + 3$	1. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
2. Solve the nonlinear system. $x^2 + y^2 = 26$ $y = x + 4$	2. 10 pts
3. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x+7)(x+8)}{(x-10)(x+7)}$	3. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts

Test continues on the other side of the sheet.

MATH 101 **72324**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Which pair satisfies h(x) = f(g(x))?

$h(x) = (8x + 8)^2 + 9$

$A.\quad f(x) = 8x + 8,\ g(x) = x^2 + 9$

$B.\quad f(x) = 8x^2 + 9,\ g(x) = x + 8$

$C.\quad f(x) = x^2 + 9,\ g(x) = 8x + 8$

$D.\quad f(x) = x^2 + 8,\ g(x) = 8x + 9$

4.

10 pts

5. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end.

$f(x) = 6\sqrt{x + 4} - 42$

5. (a)

2 pts

(b)

2 pts

(c)

1 pts

(d)

1 pts

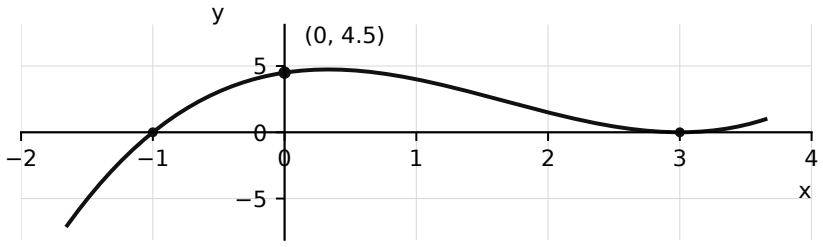
(e)

3 pts

(f)

1 pts

6. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form.



6.

10 pts

Test continues on the next sheet.

MATH 101 **72324**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

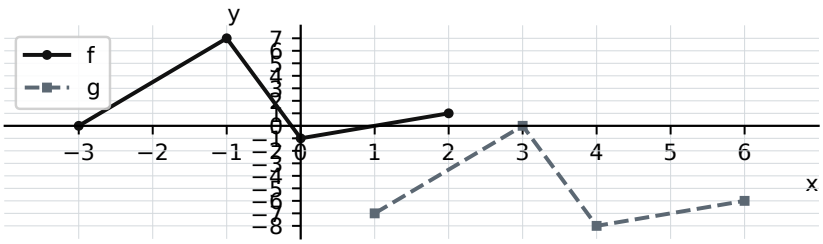
7. The solid graph is f and the dashed graph is g. Select the expression for g(x).

A. $g(x) = f(x+4) + 7$

B. $g(x) = f(x-4) + 7$

C. $g(x) = f(x+4) - 7$

D. $g(x) = f(x-4) - 7$



7.

10 pts

8. Evaluate exactly. Here log means base 10.

(*a*) $\log(0.25) + \log(400)$

(*b*) $\frac{\log_{11}(512)}{\log_{11}(64)}$

8. (a)

5 pts

(b)

5 pts

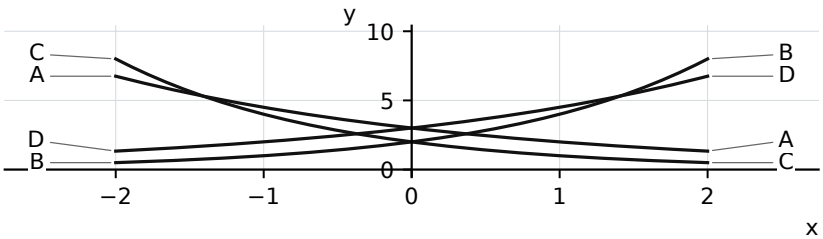
9. Match each function to graph A, B, C, or D. Each letter is used once.

(*a*) $2(\frac{1}{2})^x$

(*b*) $3(\frac{2}{3})^x$

(*c*) $3(\frac{3}{2})^x$

(*d*) $2(2)^x$



9. (a)

3 pts

(b)

2 pts

(c)

3 pts

(d)

2 pts

10. Find all zeros.

$p(x) = x^3 - 2x^2 - 4x + 3$

10.

10 pts

72324

TEST PAGE 1

1. (a)
$(7, \infty)$ 2 pts

(b)
$(-\infty, \infty)$ 2 pts

(c)
$x = 7$ 1 pts

(d)
$(\frac{57}{8}, 0)$ 1 pts

(e)
$x \to 7^+ : f(x) \to -\infty$ 2 pts

(f)
$x \to \infty : f(x) \to \infty$ 2 pts

2.
$(1, 5); (-5, -1)$
10 pts

3. (a)
$\mathbb{R} \setminus \{-7, 10\}$ 2 pts

(b)
$(-7, -\frac{1}{17})$ 2 pts

(c)
$x = 10$ 1 pts

(d)
$y = 1$ 1 pts

(e)
$(-8, 0)$ 2 pts

(f)
$(0, -\frac{4}{5})$ 2 pts

TEST PAGE 2

4.
C
10 pts

5. (a)
$[-4, \infty)$ 2 pts

(b)
$[-42, \infty)$ 2 pts

(c)
$(45, 0)$ 1 pts

(d)
$(0, -30)$ 1 pts

(e)
$x \to \infty : f(x) \to \infty$ 3 pts

(f)
$x \to -4^+ : f(x) \to -42$ 1 pts

6.
$\frac{1}{2}(x+1)(x-3)^2$
10 pts

TEST PAGE 3

7.
D
10 pts

8. (a)
2 5 pts

(b)
$\frac{3}{2}$ 5 pts

9. (a)
C
3 pts

(b)
A
2 pts

(c)
D
3 pts

(d)
B
2 pts

10.
$x = 3, -\frac{1}{2} + \frac{\sqrt{5}}{2}, -\frac{\sqrt{5}}{2} - \frac{1}{2}$ 10 pts

MATH 101 **80990**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find all zeros.

$p(x) = x^3 + 5x^2 + 7x + 2$

1.

10 pts

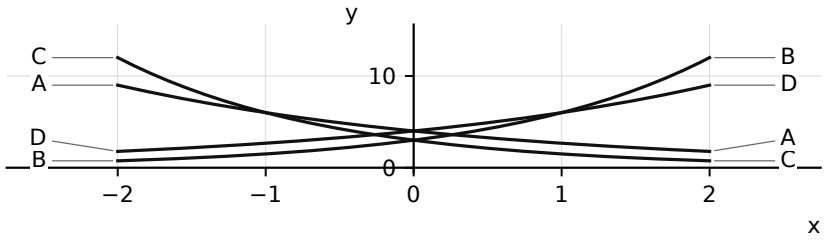
2. Match each function to graph A, B, C, or D. Each letter is used once.

$(a)\quad 3(\frac{1}{2})^x$

$(b)\quad 4(\frac{2}{3})^x$

$(c)\quad 4(\frac{3}{2})^x$

$(d)\quad 3(2)^x$



2. (a)

3 pts

(b)

2 pts

(c)

3 pts

(d)

2 pts

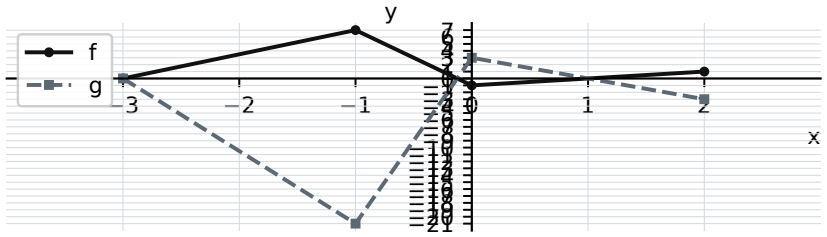
3. The solid graph is f and the dashed graph is g. Select the expression for g(x).

$A.\quad g(x) = f(x) - 3$

$B.\quad g(x) = 3f(x)$

$C.\quad g(x) = f(-3x)$

$D.\quad g(x) = -3f(x)$



3.

10 pts

Test continues on the other side of the sheet.

MATH 101 **80990**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x+7)(x+8)}{(x-10)(x+7)}$	4. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
5. Which pair satisfies h(x) = f(g(x))? $h(x) = (8x+8)^2 + 9$ $A.\quad f(x) = 8x+8,\ g(x) = x^2+9$ $B.\quad f(x) = 8x^2+9,\ g(x) = x+8$ $C.\quad f(x) = x^2+9,\ g(x) = 8x+8$ $D.\quad f(x) = x^2+8,\ g(x) = 8x+9$	5. 10 pts
6. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_3(x-7) + 3$	6. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts

Test continues on the next sheet.

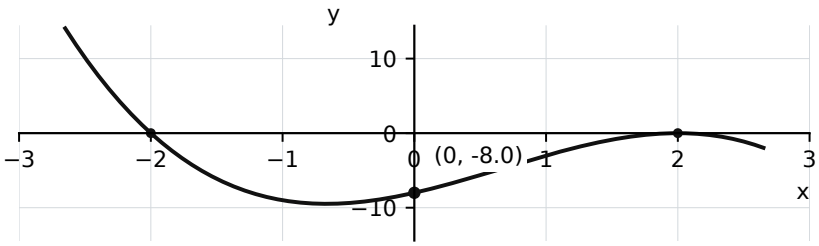
MATH 101 **80990**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

7. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 6\sqrt{x+9} - 42$	7. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts
8. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form. 	8. 10 pts
9. Evaluate exactly. Here log means base 10. $(a)\quad \log(0.8) + \log(1250)$ $(b)\quad \frac{\log_{11}(512)}{\log_{11}(64)}$	9. (a) 5 pts (b) 5 pts
10. Solve exactly. $\sqrt{x+5} = x - 5$	10. 10 pts

80990

TEST PAGE 1

1.

$x = -2,\ -\frac{3}{2} - \frac{\sqrt{5}}{2},\ -\frac{3}{2} + \frac{\sqrt{5}}{2}$

10 pts

2. (a)

C

3 pts

(b)

A

2 pts

(c)

D

3 pts

(d)

B

2 pts

3.

D

10 pts

TEST PAGE 2

4. (a)

$\mathbb{R} \setminus \{-7, 10\}$ 2 pts

(b)

$(-7,\ -\frac{1}{17})$ 2 pts

(c)

$x = 10$ 1 pts

(d)

$y = 1$ 1 pts

(e)

$(-8, 0)$ 2 pts

(f)

$(0,\ -\frac{4}{5})$ 2 pts

5.

C

10 pts

6. (a)

$(7, \infty)$ 2 pts

(b)

$(-\infty, \infty)$ 2 pts

(c)

$x = 7$ 1 pts

(d)

$(\frac{190}{27}, 0)$ 1 pts

(e)

$x \to 7^+ : f(x) \to -\infty$ 2 pts

(f)

$x \to \infty : f(x) \to \infty$ 2 pts

TEST PAGE 3

7. (a)

$[-9, \infty)$ 2 pts

(b)

$[-42, \infty)$ 2 pts

(c)

$(40, 0)$ 1 pts

(d)

$(0,\ -24)$ 1 pts

(e)

$x \to \infty : f(x) \to \infty$ 3 pts

(f)

$x \to -9^+ : f(x) \to -42$ 1 pts

8.

$-1(x+2)(x-2)^2$

10 pts

9. (a)

3 5 pts

(b)

$\frac{3}{2}$ 5 pts

10.

$x = \frac{\sqrt{41}}{2} + \frac{11}{2}$

10 pts

MATH 101 **27549**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x+7)(x+8)}{(x-10)(x+7)}$	1. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
2. Which pair satisfies h(x) = f(g(x))? $h(x) = (8x+8)^2 + 9$ $A.\quad f(x) = 8x+8,\ g(x) = x^2+9$ $B.\quad f(x) = 8x^2+9,\ g(x) = x+8$ $C.\quad f(x) = x^2+9,\ g(x) = 8x+8$ $D.\quad f(x) = x^2+8,\ g(x) = 8x+9$	2. 10 pts
3. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 6\sqrt{x+16} - 42$	3. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts

Test continues on the other side of the sheet.

MATH 101 **27549**

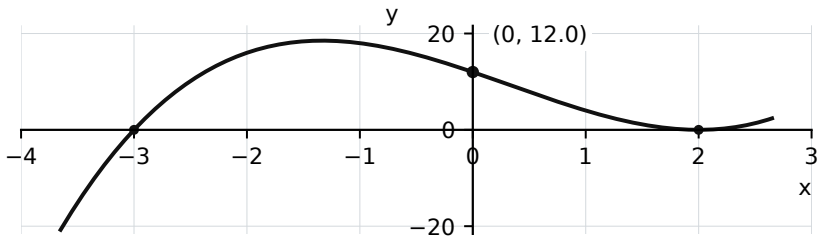
Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form.



4.

10 pts

5. Evaluate exactly. Here log means base 10.

(*a*) $\log(0.05) + \log(200000)$

(*b*) $\frac{\log_{11}(512)}{\log_{11}(64)}$

5. (a)

5 pts

(b)

5 pts

6. Find all zeros.

$p(x) = x^3 + 3x^2 - 2$

6.

10 pts

7. Solve exactly.

$8^{x-4} = 4^{x+4}$

7.

10 pts

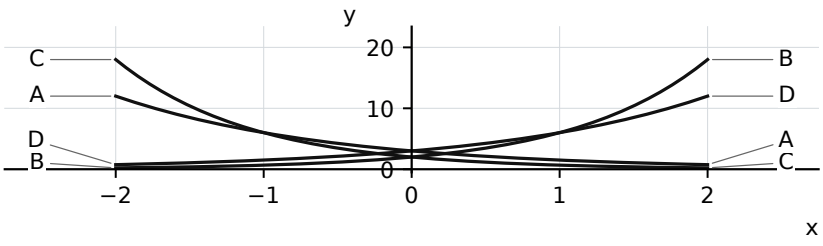
8. Match each function to graph A, B, C, or D. Each letter is used once.

(*a*) $2(\frac{1}{3})^x$

(*b*) $3(\frac{1}{2})^x$

(*c*) $3(2)^x$

(*d*) $2(3)^x$



8. (a)

3 pts

(b)

2 pts

(c)

3 pts

(d)

2 pts

Test continues on the next sheet.

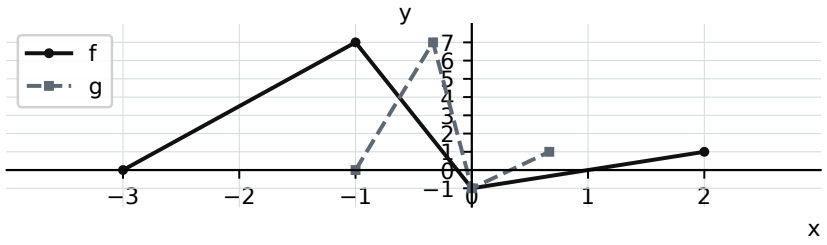
MATH 101 **27549**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

9. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_4(x-7)+3$	9. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
10. The solid graph is f and the dashed graph is g. Select the expression for g(x). $A.\quad g(x) = f(x-3)$ $B.\quad g(x) = f(x/3)$ $C.\quad g(x) = 3f(x)$ $D.\quad g(x) = f(3x)$ 	10. 10 pts

27549

TEST PAGE 1

1. (a)
$\mathbb{R} \setminus \{-7, 10\}$ 2 pts

(b)
$(-7, -\frac{1}{17})$ 2 pts

(c)
$x = 10$ 1 pts

(d)
$y = 1$ 1 pts

(e)
$(-8, 0)$ 2 pts

(f)
$(0, -\frac{4}{5})$ 2 pts

2.
C

10 pts

3. (a)
$[-16, \infty)$ 2 pts

(b)
$[-42, \infty)$ 2 pts

(c)
$(33, 0)$ 1 pts

(d)
$(0, -18)$ 1 pts

(e)
$x \to \infty : f(x) \to \infty$ 3 pts

(f)
$x \to -16^+ : f(x) \to -42$ 1 pts

TEST PAGE 2

4.
$1(x+3)(x-2)^2$

10 pts

5. (a)
4 5 pts

(b)
$\frac{3}{2}$ 5 pts

6.
$x = -1, -1+\sqrt{3}, -\sqrt{3}-1$

10 pts

7.
$x = 20$

10 pts

8. (a)
C

3 pts

(b)
A

2 pts

(c)
D

3 pts

(d)
B

2 pts

TEST PAGE 3

9. (a)
$(7, \infty)$ 2 pts

(b)
$(-\infty, \infty)$ 2 pts

(c)
$x = 7$ 1 pts

(d)
$(\frac{449}{64}, 0)$ 1 pts

(e)
$x \to 7^+ : f(x) \to -\infty$ 2 pts

(f)
$x \to \infty : f(x) \to \infty$ 2 pts

10.
D

10 pts

MATH 101 **40669**

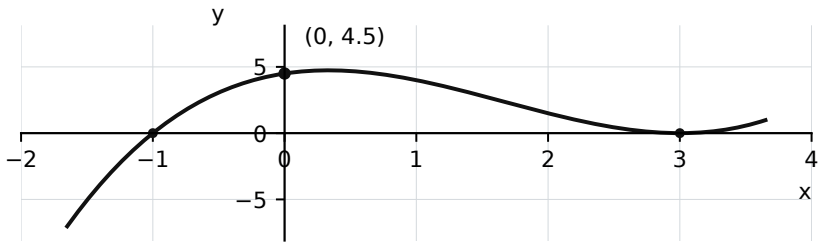
Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form.



1.

10 pts

2. Solve exactly.

$\log_4(x-4) - \log_4(x-8) = 1$

2.

10 pts

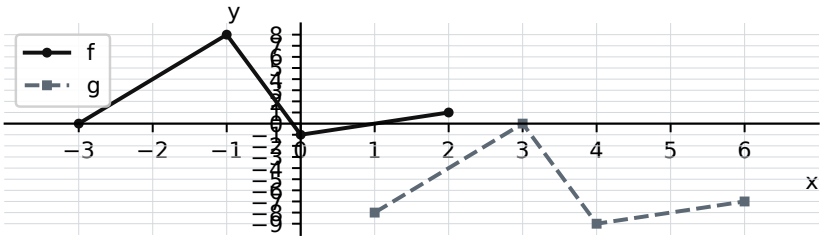
3. The solid graph is f and the dashed graph is g. Select the expression for g(x).

A. $g(x) = f(x+4) - 8$

B. $g(x) = f(x-4) + 8$

C. $g(x) = f(x+4) + 8$

D. $g(x) = f(x-4) - 8$



3.

10 pts

4. Evaluate exactly. Here log means base 10.

(*a*) $\log(0.25) + \log(400)$

(*b*) $\dfrac{\log_{12}(729)}{\log_{12}(81)}$

4. (a)

5 pts

(b)

5 pts

Test continues on the other side of the sheet.

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

5. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x+8)(x+9)}{(x-11)(x+8)}$	5. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
6. Which pair satisfies h(x) = f(g(x))? $h(x) = (9x+9)^2 + 10$ $A.\quad f(x) = 9x^2 + 10,\ g(x) = x + 9$ $B.\quad f(x) = x^2 + 9,\ g(x) = 9x + 10$ $C.\quad f(x) = x^2 + 10,\ g(x) = 9x + 9$ $D.\quad f(x) = 9x + 9,\ g(x) = x^2 + 10$	6. 10 pts
7. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 7\sqrt{x+4} - 56$	7. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts

Test continues on the next sheet.

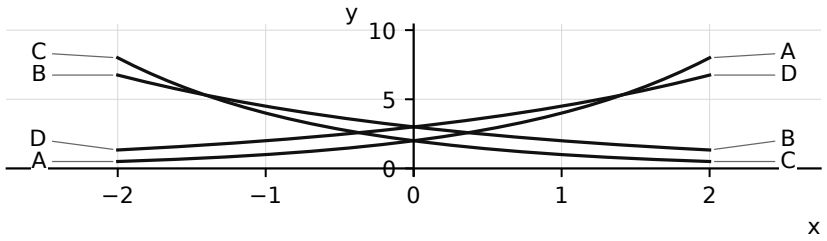
MATH 101 **40669**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

8. Find all zeros. $p(x) = x^3 - 2x^2 - 4x + 3$	8. 10 pts
9. Match each function to graph A, B, C, or D. Each letter is used once. $(a)\quad 2(\frac{1}{2})^x$ $(b)\quad 2(2)^x$ $(c)\quad 3(\frac{3}{2})^x$ $(d)\quad 3(\frac{2}{3})^x$ 	9. (a) 3 pts (b) 2 pts (c) 3 pts (d) 2 pts
10. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_2(x - 8) + 1$	10. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts

40669

TEST PAGE 1

1.

$\frac{1}{2}(x+1)(x-3)^2$

10 pts

2.

$x = \frac{28}{3}$

10 pts

3.

D

10 pts

4. (a)

2

5 pts

(b)

$\frac{3}{2}$

5 pts

TEST PAGE 2

5. (a)

$\mathbb{R} \setminus \{-8, 11\}$

2 pts

(b)

$(-8, -\frac{1}{19})$

2 pts

(c)

$x = 11$

1 pts

(d)

$y = 1$

1 pts

(e)

$(-9, 0)$

2 pts

(f)

$(0, -\frac{9}{11})$

2 pts

6.

C

10 pts

7. (a)

$[-4, \infty)$

2 pts

(b)

$[-56, \infty)$

2 pts

(c)

$(60, 0)$

1 pts

(d)

$(0, -42)$

1 pts

(e)

$x \to \infty : f(x) \to \infty$

3 pts

(f)

$x \to -4^+ : f(x) \to -56$

1 pts

TEST PAGE 3

8.

$x = 3, -\frac{1}{2} + \frac{\sqrt{5}}{2}, -\frac{\sqrt{5}}{2} - \frac{1}{2}$

10 pts

9. (a)

C

3 pts

(b)

A

2 pts

(c)

D

3 pts

(d)

B

2 pts

10. (a)

$(8, \infty)$

2 pts

(b)

$(-\infty, \infty)$

2 pts

(c)

$x = 8$

1 pts

(d)

$(\frac{17}{2}, 0)$

1 pts

(e)

$x \to 8^+ : f(x) \to -\infty$

2 pts

(f)

$x \to \infty : f(x) \to \infty$

2 pts

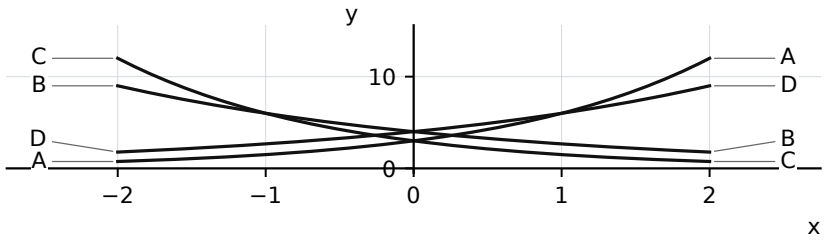
MATH 101 **58687**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 7\sqrt{x+9} - 56$	1. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts
2. Solve the nonlinear system. $y = x^2$ $y = x + 2$	2. 10 pts
3. Evaluate exactly. Here log means base 10. $(a)\quad \log(0.8) + \log(1250)$ $(b)\quad \frac{\log_{12}(729)}{\log_{12}(81)}$	3. (a) 5 pts (b) 5 pts
4. Match each function to graph A, B, C, or D. Each letter is used once. $(a)\quad 3(\frac{1}{2})^x$ $(b)\quad 3(2)^x$ $(c)\quad 4(\frac{3}{2})^x$ $(d)\quad 4(\frac{2}{3})^x$ 	4. (a) 3 pts (b) 2 pts (c) 3 pts (d) 2 pts

Test continues on the other side of the sheet.

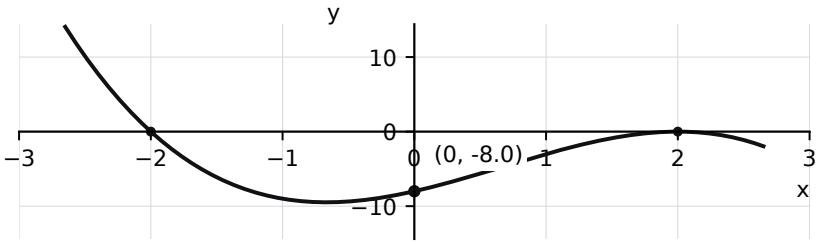
MATH 101 **58687**

Final Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

5. Find all zeros. $p(x) = x^3 + 5x^2 + 7x + 2$	5. 10 pts
6. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form. 	6. 10 pts
7. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x+8)(x+9)}{(x-11)(x+8)}$	7. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts

Test continues on the next sheet.

MATH 101 **58687**

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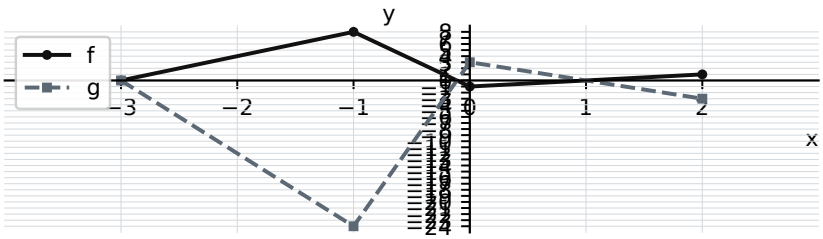
8. The solid graph is f and the dashed graph is g. Select the expression for g(x).

$A.\quad g(x) = f(-3x)$

$B.\quad g(x) = 3f(x)$

$C.\quad g(x) = f(x) - 3$

$D.\quad g(x) = -3f(x)$



8.

10 pts

9. Which pair satisfies h(x) = f(g(x))?

$h(x) = (9x + 9)^2 + 10$

$A.\quad f(x) = 9x^2 + 10,\ g(x) = x + 9$

$B.\quad f(x) = x^2 + 9,\ g(x) = 9x + 10$

$C.\quad f(x) = x^2 + 10,\ g(x) = 9x + 9$

$D.\quad f(x) = 9x + 9,\ g(x) = x^2 + 10$

9.

10 pts

10. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end.

$f(x) = \log_3(x - 8) + 1$

10. (a)

2 pts

(b)

2 pts

(c)

1 pts

(d)

1 pts

(e)

2 pts

(f)

2 pts

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TEST PAGE 1

1. (a)

$[-9, \infty)$ — 2 pts

(b)

$[-56, \infty)$ — 2 pts

(c)

$(55, 0)$ — 1 pts

(d)

$(0, -35)$ — 1 pts

(e)

$x \to \infty : f(x) \to \infty$ — 3 pts

(f)

$x \to -9^+ : f(x) \to -56$ — 1 pts

2.

$(-1, 1); (2, 4)$

10 pts

3. (a)

3 — 5 pts

(b)

$\frac{3}{2}$ — 5 pts

4. (a)

C

3 pts

(b)

A

2 pts

(c)

D

3 pts

(d)

B

2 pts

TEST PAGE 2

5.

$x = -2, -\frac{3}{2} - \frac{\sqrt{5}}{2}, -\frac{3}{2} + \frac{\sqrt{5}}{2}$

10 pts

6.

$-1(x+2)(x-2)^2$

10 pts

7. (a)

$\mathbb{R} \setminus \{-8, 11\}$ — 2 pts

(b)

$(-8, -\frac{1}{19})$ — 2 pts

(c)

$x = 11$ — 1 pts

(d)

$y = 1$ — 1 pts

(e)

$(-9, 0)$ — 2 pts

(f)

$(0, -\frac{9}{11})$ — 2 pts

TEST PAGE 3

8.

D

10 pts

9.

C

10 pts

10. (a)

$(8, \infty)$ — 2 pts

(b)

$(-\infty, \infty)$ — 2 pts

(c)

$x = 8$ — 1 pts

(d)

$(\frac{25}{3}, 0)$ — 1 pts

(e)

$x \to 8^+ : f(x) \to -\infty$ — 2 pts

(f)

$x \to \infty : f(x) \to \infty$ — 2 pts